# Axiomatic Systems Programming: Structural Type Theory and Exact Arithmetic in C++23

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April 14, 2026

**Abstract**

This paper introduces `dedekind`, a `C++23` library designed to embed formal mathematical structures directly within the language's type system as first-class, verified specifications. We address the performance-productivity gap in computational science—often characterized as the "two-language problem"—by leveraging modern `C++` features, including modules, concepts, and non-type template parameters (NTTP), to implement a structural type theory. Unlike traditional object-oriented frameworks that rely on nominal inheritance, `dedekind` reifies mathematical entities as positions within a categorical system of relations.

We demonstrate the utility of this approach through a formal construction of the continuum $\mathbb{R}$ via Dedekind cuts, enabling the exact resolution of transcendental numbers without floating-point overhead. Furthermore, we show how hoisting mereological and algebraic invariants into type signatures allows the `LLVM` backend to perform aggressive structural pruning, such as collapsing contradictory set-builder predicates into zero-instruction machine code. By addressing language-level constraints through the introduction of the "Highway Notation" for Kleisli composition, this work illustrates that integrating formal mathematical rigor into systems programming can yield zero-overhead abstractions suitable for high-performance symbolic computing. We propose this methodology as a nascent paradigm for *Axiomatic Systems Programming*, where the compiler serves as a formal verification engine for mathematical truth.

# Contents

# Chapter 1

# Introduction

The development of modern computational science, particularly in data science and Large Language Model (LLM) engineering, is constrained by a persistent performance-productivity gap known as the *two-language problem*. In this paradigm, researchers utilize high-level symbolic environments like `Python` for rapid prototyping and expressive reasoning, while relying on low-level languages like C++ to handle execution backends. While tools such as `PyTorch` [1] and `NumPy` [2] offer elegant interfaces, they typically treat mathematical logic as a runtime implementation detail. This decoupling forces a manual translation layer between the high-level mathematical intent and the underlying hardware execution, leading to significant overhead in both developer time and system latency.

This duality is especially relevant in the context of LLM development and large-scale symbolic computing. In these fields, the ability to maintain the structural integrity of a mathematical model—such as the exactness of a tensor operation or the logical consistency of a set-builder predicate—is often lost during the transition to the C++ backend. Traditional integration methods, such as `ctypes` or even modern binding tools like `nanobind` [3], bridge the binary gap but do not bridge the *semantic* gap. They allow Python to call C++ functions, but they do not allow the C++ compiler to reason about the high-level mathematical invariants defined in the symbolic layer.

Recent advances in systems programming have begun to challenge the historical divide between low-level performance and high-level abstraction. The `Rust` programming language, for instance, has gained significant traction by providing a "safety-first" approach through its affine type system and trait-based polymorphism. However, while Rust excels at memory safety and nominal constraints, its type system often operates as a "black box" that can obscure the structural transparency required to hoist complex mathematical invariants into the heart of the optimization pipeline.

In contrast, the evolution of C++ toward the `C++23` standard [4] has introduced a suite of features—notably `concepts`, `consteval`, and non-type template parameters (NTTP)—that enable a form of *structural typing*. Unlike traditional object-oriented hierarchies, these tools allow developers to define "mathematical species" based on their internal requirements and relational properties rather than their inherited names. As explored by Čukić [5], modern C++ is now sufficiently advanced to simulate, and in some cases surpass, the expressive power of pure functional programming languages. By treating types not merely as memory layouts but as compile-time predicates, C++ can now reify abstract "notions of computation" directly into the machine code. This shift allows for the embedding of category-theoretic patterns, such as monads and functors, which serve as the formal foundation for structured side-effects and composition [6, 7].

The `dedekind` library specifically adopts the framework of *combinatorial species* introduced by Joyal [8, 9]. By treating mathematical structures as "species"—mappings that describe how a structure is placed upon a set of labels—we can reify abstract species (such as rings, fields,

or even the continuum itself) as C++ template specifications. This "white-box" structural transparency allows the compiler to reason about the properties of a type (e.g., its commutativity or its completeness) purely by inspecting its requirements, much like McLarty's observation that "numbers can be just what they have to" within a categorical framework [10].

This shift toward compile-time verification mirrors the transition from substance-based to structuralist ontologies. While traditional object-oriented systems rely on **nominal typing**—where an object's identity is tied to its named inheritance—`dedekind` adopts a **structural type theory**. This follows the mathematical structuralism of Awodey [7], where objects are defined by their roles and relations rather than an internal "essence." While "duck typing" served as a dynamic precursor to this idea, C++ Concepts allow for its formal, static verification. As Juliet observes, "a rose by any other name would smell as sweet" [11]; we prioritize the "scent" (structural invariants) over the "lineage" (nominal inheritance), as shown in Listing 1.1.

```cpp
// Nominal: Recognition by lineage
struct Flower { virtual bool smells_sweet() const = 0; };
struct Rose : public Flower { ... };

// Structural: Recognition by scent (C++23)
template <typename T>
concept SmellsSweet = requires(T t) {
    { t.scent() } -> std::same_as<Sweetness>;
};

// In dedekind, a 'Briar' is a 'Rose' if it satisfies the scent.
static_assert(SmellsSweet<Briar>);
```

Listing 1.1: The Juliet Posture: Nominal vs. Structural Recognition.

**Two-Tier Concept Design: Bridging Signature and Semantics.** A critical architectural innovation introduced in PR #111 is the **Two-Tier Concept Design**, which addresses a fundamental challenge in compile-time mathematical verification: the undecidability of algebraic laws. While syntactic properties ("Does this type support a `meet` operation?") are fully decidable via C++23 Concepts, semantic properties ("Is the `meet` operation associative?") are generally undecidable for arbitrary types. Rather than making impossible demands on the type system, `dedekind` partitions verification into two complementary tiers:

1. **Tier 1 (Structural Verification)**: Concepts that check only the *signature*—the existence of required operations and their type compatibility. These have zero runtime cost and are suitable for generic programming where correctness follows from structure alone.

2. **Tier 2 (Certified Verification)**: Extended concepts that require explicit trait witnesses for algebraic laws (e.g., commutativity, associativity). These rely on a registry of well-known species (integers, floating-point types) and allow specialization for user-defined types via trait witnesses. Crucially, final validation of laws occurs in the test suite, where properties are verified at runtime against concrete instantiations.

This design reflects a principle we term *Semantic Honesty*: the library documents and verifies which laws hold for which species, and users (via tests and CI) are responsible for validating these properties. This yields a practical balance between the theoretical rigor of category theory and the pragmatic constraints of compile-time verification.

By treating mathematical structures as **combinatorial species** [8], we allow the compiler to reason about abstract properties purely by inspecting type requirements. This "white-box" transparency enables the compiler to identify a type as a representation of $\mathbb{R}$ if it satisfies the requirements of a Dedekind-complete ordered field, regardless of its name. These mappings and their categorical implications are summarized in Table 5.3 and Table 5.4.

```cpp
template <typename Universe, auto Predicate>
struct Set {
    using type = Universe;
    static constexpr auto p = Predicate;
};

// Mereological composition (Intersection)
template <typename S1, typename S2>
using Intersection = Set<typename S1::type, [](auto x) {
    return S1::p(x) && S2::p(x);
}>;
```

Listing 1.2: A simplified illustration of a set intersection implementation in C++ in the structuralist, mereological style.

In practice, this allows us to move beyond the "black-box" constraints of nominal traits. For instance, while a nominal system might require a type to explicitly `impl Real`, a structural system in C++ can identify a type as a representation of $\mathbb{R}$ if it satisfies the structural requirements of a Dedekind-complete ordered field. By hoisting these mereological invariants into the type signature, we enable the LLVM backend to perform aggressive structural pruning. As noted in the *n-Category Café* [12], this categorical perspective treats the "logic" of the system and the "computation" of the system as dual aspects of the same structure—a principle we aim to exploit. For example, to ensure that contradictory set-builder predicates collapse into zero-instruction machine code (see Listing 1.2).

```cpp
using
 = Set<
, [](auto x) { return x > 0.0; }>;
using
 = Set<
, [](auto x) { return false; }>;
// Intersection with
 is symbolically reduced
using Result = Intersection<
,
>;
// Proven at compile-time: Result is the empty set
static_assert(Result::p(5.0) == false);
```

Listing 1.3: Compile-time reduction of contradictory set intersections.

The utility of this structuralist approach is best illustrated through two distinct computational advantages. First, the library enables aggressive structural pruning at compile-time. By reifying set operations as type-level compositions, the compiler can symbolically resolve the relationship between predicates. For instance, the intersection of two contradictory sets—such as the set of positive reals and the set of negative reals—can be identified as an empty set during translation to LLVM Intermediate Representation (IR). This allows the backend to prune entire execution branches, effectively collapsing complex symbolic logic into zero-instruction machine code.

```cpp
// Representing e via the series predicate:
// q < sum_{n=0}^N 1/n! for some N
using e = Set<ℚ, [](ℚ q) {
    // Symbolic check: ∃ N such that q < partial_sum(N)
    // In dedekind, this is resolved by the compiler's
    // ability to evaluate monotonic convergence.
    return ∃_approximation(q, [](int n) {
        return factorial_reciprocal_sum(n);
    });
```

```
}>;

// The compiler resolves this comparison at translation time
static_assert(e::p({271, 100}) == true); // 2.71 < e
```

Listing 1.4: Illustration of the symbolic construction of $e$ as an intensional set (the lower cut $L \subset \mathbb{Q}$).

Furthermore, `dedekind` provides the ability to model intensional, transfinite sets through a purely symbolic approach. Rather than representing a collection as an extensional buffer of values, we define it by its membership condition. This is most powerfully demonstrated in our construction of the continuum $\mathbb{R}$ via Dedekind cuts. By representing a real number as a compile-time coordinate—a partition of the rational numbers $\mathbb{Q}$ —we enable the exact resolution of transcendental values without the rounding errors or overhead of standard floating-point arithmetic [13]. As shown in Listing 1.5, comparing $e$ and $\pi$ becomes a compile-time verification of their respective partitions.

```
// Defined as the lower cut of their respective series/properties
using e   = Set<ℚ, [](auto q) { return ∃_lower_bound(q, e_series); }>;
using π   = Set<ℚ, [](auto q) { return ∃_lower_bound(q, π_series); }>;

/**
 * Structural comparison: Is e < pi?
 * This is true if there exists a rational 'r' such that:
 * r is in the lower cut of pi, but NOT in the lower cut of e.
 */
static_assert(e < π);
```

Listing 1.5: Illustration of the comparison of the transcendental constants $e$ and $\pi$ via their Dedekind partitions.

Together, these features bridge the gap between the declarative ease of symbolic algebra and the mechanical sympathy required for modern high-performance computing [14]. By allowing the C++ compiler itself to understand the symbolic logic of the Continuum, `dedekind` ensures that high-level mathematical invariants are preserved from specification to machine code, effectively treating the compiler as a formal proof assistant for systems-level performance.

# Chapter 2

# Design and Methodology

The methodology of `dedekind` is predicated on the translation of formal mathematical structures into a stratified registry of C++23 modules. This stratification allows for a "bottom-up" synthesis of complex mathematical objects, where each level serves as a verified foundation for the next. By design, the flow of compilation mirrors a pedagogical progression: foundational, general-purpose modules are verified first, providing an immutable basis for the specialized and practical abstractions that follow.

Table 2.1: The Dedekind Atomic Floor: Primary Formal Correspondences

| **Mathematical Concept** | **C++23 Implementation** | **Structural `concept`** |
|---|---|---|
| *Morphic Foundations* | | |
| Morphism ($f : A \to B$) | `std::invoke_result_t` | `IsArrow` |
| Terminal Object (1) | `std::monostate` | `IsTerminalObject` |
| Initial Object (0) | `std::nullptr_t` | `IsInitialObject` |
| *Structural Relations* | | |
| Parthood ($\sqsubseteq$) | `operator<=` | `IsPartOf` |
| Subobject Classifier ($\Omega$) | `enum class Omega` | `IsLogicalSpecies` |
| *Universal Constructions* | | |
| Product ($A \times B$) | `std::pair` | `IsProduct` |
| Coproduct ($A + B$) | `std::variant` | `IsCoproduct` |

## 2.1 Instrumented Model Validation: The CI Pipeline as an Independent Auditor

While formal verification in languages like Agda or Coq typically relies on static type-level proofs that are erased during compilation, the `dedekind` framework adopts a methodology of *Instrumented Model Validation*. By leveraging the LLVM profiling infrastructure, we provide an empirical *Runtime Witness* for our categorical primitives.

This approach serves as a deterministic evolution of the property-based testing tradition pioneered by QuickCheck [15]. Where QuickCheck utilizes stochastic search to falsify algebraic properties in Haskell, `dedekind` utilizes the CI/CD pipeline, which exercises `static_assert` (evaluated at compile-time) combined with `Catch2` tests (executed after unit compilation) to ensure that every structural identity is physically materialized in the binary. By enforcing high levels of patch coverage on new ontological partitions, we provide a "Physical Proof" in the form of *Binary Module Interface* (BMI) files that the abstract category theory—often treated as a ghost in the machine—is actually traversed by the hardware. This aligns with the "Real World"

functional programming philosophy advocated by O'Sullivan et al. [16], where formal rigor is balanced with the practical necessity of observable, high-performance execution.

Further empirical evidence for the soundness of the logic is gathered by forcing all code changes through a pull request review with an AI reviewer (GitHub Copilot, in this case). Here, the reviewer is instructed to compare the logic of both the code and the test suite against literature references embedded in the code through doxygen comments.

This methodology is physically enforced by a strictly layered directed acyclic graph (DAG) of the system's build order. By carefully selecting C++ module partitions and build-step boundaries, we translate the dependencies of formal mathematics into a sequence of verified compilation units. This architecture mimics the pedagogical progression of a mathematical textbook, where the ontology matures from foundational axioms to complex theorems through a series of verified proofs. By isolating these prerequisites into a 'category' module, we leverage what Wadler termed "Theorems for Free!" [17]: once a structural identity is established and cached as a Binary Module Interface (BMI), its properties are inherited by all downstream partitions as an immutable "cached truth." This ensures that the compiler's proof-search is grounded in a physical hierarchy, preventing circular reasoning and ensuring that the structural integrity of the model is a prerequisite for successful linking. Crucially, the CI/CD methodology provides an independent, physical verification of this progression; by requiring every node in the build graph to pass its respective model validation before the pipeline can proceed, we ensure that the "textbook" derivation is not merely a logical possibility, but a materialized reality across the entire module graph.

### 2.1.1 The Compiler as a Verification and Optimization Engine

While functional languages like Haskell are often viewed as the natural home for Category Theory, Milewski [18] demonstrated that C++ is equally capable of modeling complex categorical morphisms. `dedekind` builds upon this bridge by utilizing C++23 features to move from mere simulation to compile-time verification.

By applying `static_assert` against structural concepts, we transform the compilation process into a formal verification phase. This provides the `LLVM` toolchain with the semantic context required for "structural pruning": by hoisting mereological relations into the type signature, the backend can identify logical contradictions—such as an empty set defined by contradictory predicates—and prune them into zero-overhead machine instructions. This "correctness-by-construction" methodology ensures that the structural integrity of the mathematical model is verified and optimized before the final binary is generated.

### 2.1.2 Ontological Decidability and the Build Graph

To mitigate the "template tax" and ensure the practical decidability of the ontology, we utilize a directed acyclic build graph (see Figure 2.1.2). This directed acyclic graph mirrors the formal structure of a mathematical proof: *Axiom* $\rightarrow$ *Theorem* $\rightarrow$ *Proof*. While the underlying mathematical nature of certain propositions remains inherently undecidable, `dedekind` provides structural guardrails by isolating foundational axioms into up-stream module partitions. The resulting BMI acts as the "cached truth" established in the previous section.

This physical stratification prohibits circular reasoning at the compiler level as a partition can only reference a `concept` which was previously defined. Subsequent partitions then import these BMIs as established, immutable lemmata, allowing the compiler to focus on the synthesis of higher-order laws with a greatly reduced risk of recursive definition or redundant re-instantiation. While the module graph serves as an external meta-logical constraint ensuring build-time termination, we later introduce a pluggable *Subobject Classifier* ($\Omega$) in Section 3.2.2, which internalizes the notion of undecidability within the C++ type system itself.

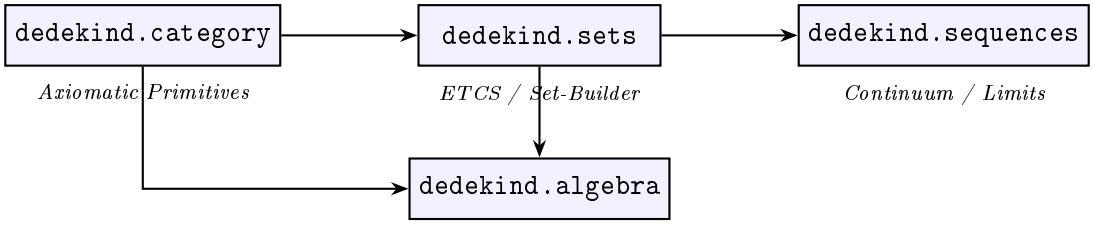


Figure 2.1: Module build dependencies in the `dedekind` library.

For the remainder of the implementation section, we simply follow build order of the system. We will annotate and document the specific modules and partitions as they are defined and validated, beginning with the foundational internal partitions of the `dedekind.category` module.

In this approach, the structure of the paper is intended to be isomorphic to the CI/CD pipeline: the text serves as a formal documentation of the build, while the build, in turn, provides a physical validation of the claims made within the paper. By the time the reader reaches the final sections, the library's 'textbook' derivation will have been fully materialized as a verified C++ library object traversed by the underlying hardware.

# Chapter 3

# dedekind.category

## 3.1 Introduction

The `dedekind.category` module constitutes the system's "Atomic Floor," providing the foundational categorical primitives upon which the entire library rests. Following the pedagogical progression of a formal mathematical textbook, its internal build graph generates a rigid lattice of verified structural invariants—morphisms, mereology, logic, actions, monads, and an ETCS set-theoretic interface—before any numeric or domain-specific module is compiled. Every downstream partition imports from `dedekind.category` and can therefore rely on its axioms as immutable, cached Binary Module Interface (BMI) lemmata.

The key references for this module are the categorical foundations of Lawvere [19], the monadic calculus of Moggi [6], the Kleisli construction [20], and the species theory of Joyal [8].

## 3.2 Implementation

### 3.2.1 Level 0: The Algebraic Atlas

While the global build graph establishes the dependency between C++ modules, the `dedekind.category` module constitutes the system's "Atomic Floor." To maintain absolute linear order, this module is itself stratified into internal partitions. These partitions represent the maturation of mathematical species from raw mappings (`:morphism`) to structured universes (`:logic`).

#### The `:species` partition: The Algebraic Atlas

The `:species` partition serves as the foundational *Algebraic Atlas* of the library, providing a stratified registry of structural invariants for C++ machine types and standard library operators. Following the combinatorial species theory of Joyal [8, 9], we treat fundamental types not merely as passive sets of values, but as active rules for generating mathematical structure. By reifying categorical axioms—such as associativity and commutativity—as compile-time constants, the library transforms the C++ type system into a formal verification engine.

As illustrated in Table 3.1, the **Feature Cube** acts as a formal ledger mapping algebraic properties to the language's fundamental atoms. This explicit mapping allows the `dedekind` dispatcher to distinguish between idealized algebraic structures (e.g., `unsigned int` modular addition) and the messy inhabitants of the hardware, such as floating-point operations which exhibit numerical non-associativity under the IEEE 754 standard [13]. By anchoring these operators as first-class categorical entities, we enable the compiler to act as a physical guardrail, ensuring that structural integrity is verified before the final binary is generated.

Crucially, this allows for the *Honest Rejection* of hardware species that violate the laws of the intended mathematical space: while `unsigned int` addition satisfies the *Monoid* axioms, it

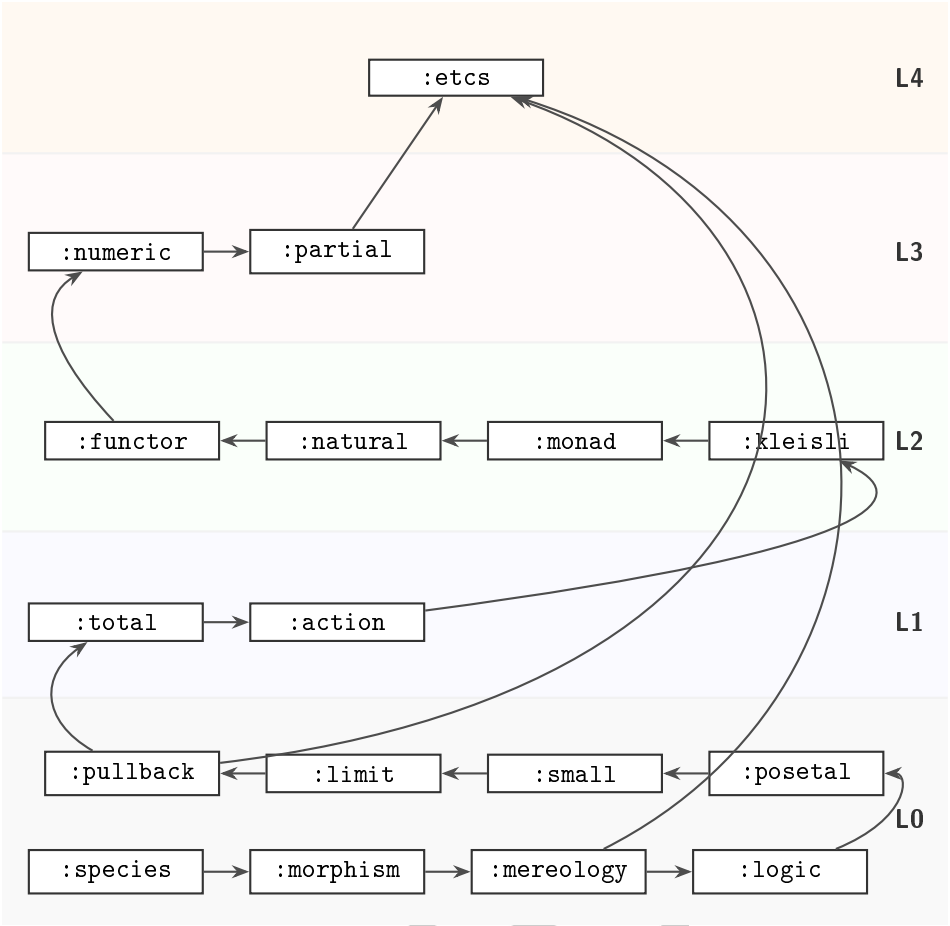


Figure 3.1: Refined Build Graph: The "Numeric" layer now serves as the base for "Partial" logic, allowing "ETCS" to remain decoupled from hardware specificities.

fails as an *Ordered Monoid* at the Lipschitz boundary due to modular wrap-around. Conversely, `signed int` is treated as a *Hazardous Ordered Monoid* where the compiler exploits UB to enforce a false sense of algebraic harmony through assumed monotonicity.

```
/**
 * @brief The Feature Cube: Categorical axioms as compile-time traits.
 * @tparam T The Species (e.g., int, double).
 * @tparam Op The Binary Operation or Relation (e.g., std::plus<>).
 */
template <typename T, typename Op>
struct SpeciesTraits {
  static constexpr bool is_associative = false;
  static constexpr bool is_commutative = false;
  static constexpr bool is_idempotent  = false;
};

// Specialization for 'Ideal' Integral Species
template <>
struct SpeciesTraits<int, std::plus<int>> {
  static constexpr bool is_associative = true;
  static constexpr bool is_commutative = true;
};

// Verification: The compiler now "knows" the algebra of the machine.
```

```
static_assert(SpeciesTraits<int, std::plus<int>>::is_associative,
              "Integrals form a valid Semigroup.");

static_assert(!SpeciesTraits<double, std::plus<double>>::is_associative,
              "Hardware Floats are non-associative; Rescue required.");
```

Listing 3.1: The Species Registry: Reifying Algebraic Invariants.

The core objective is to anchor the C++ type system as a substrate for modeling truths obtained from category theory and abstract algebra. In this structuralist setup, providing the axiomatic truths (e.g., that $a \vee b$ is commutative) or documenting their absence (e.g., $NaN$ singularities) becomes the explicit responsibility of the registry. This static grounding provides the necessary "alphabet" for the `:morphism` partition to begin defining the "language" of functional mappings.

### The `:morphism` partition: The Skeletal Arrow

**Type-Level Labelling:** We employ type-level labelling to associate categorical metadata—specifically domain and codomain species—with C++ types. By utilizing template traits and nested aliases, these labels exist strictly at compile-time. This mechanism enables the direct embedding of primitive and standard library types (e.g., `std::optional`) into formal categorical structures with zero runtime overhead and without the need for invasive wrappers.

**Morphic Reification and Signature Discovery:** While C++ lambdas are inherently opaque, anonymous types, the `:morphism` partition utilizes a *Morphic Bridge* to lift them into first-class categorical entities. By leveraging `std::invoke_result_t` and variadic template pack deduction, the library automatically extracts the domain and codomain of any callable.

The `:morphism` partition establishes the primary inhabitant of the `dedekind` topos: the **Arrow** (or Morphism). While traditional C++ development treats functions as opaque, executable blocks, this partition reifies them as formal mappings $f : A \to B$. **In this ontology, the Morphism serves as the "Atom of Composability"**—the fundamental unit from which all higher-level categorical structures are synthesized. By elevating the function object from a procedural instruction to a structural entity, we enable the library to reason about the composition and validity of transformations at compile-time, a strategy central to modern functional C++ architectures [5, 18].

A core mechanism of this partition is **Signature Discovery**. Utilizing C++23 `std::invoke_result_t` and variadic template metaprogramming, the library automatically extracts the domain and codomain of raw lambdas, function pointers, and functors. This reification process registers the callable into the categorical atlas as a first-class inhabitant, providing the "connective tissue" required for the algebraic structures in Level 1 and the functorial lifting in Level 2.

As shown in Listing 3.2, the `IsArrow` concept acts as a formal gatekeeper. It ensures that every mapping is well-formed—possessing a verifiable signature and satisfying the axioms of a morphism—before it can be used as a taxonomic brick in the broader structural hierarchy [21].

```
/** @brief Automatic inference of Morphism metadata f: Domain -> Codomain */
template <typename F, typename... Args>
struct SpeciesTraits<F, Args...> {
  using Domain   = std::tuple_element_t<0, std::tuple<Args...>>;
  using Codomain = std::invoke_result_t<F, Args...>;
  static constexpr bool is_morphism = true;
};

/** @brief Concept enforcing the integrity of the Skeletal Arrow */
template <typename F, typename... Args>
concept IsArrow = SpeciesTraits<F, Args...>::is_morphism &&
                  requires { typename SpeciesTraits<F, Args...>::Codomain; };
```

Table 3.1: The Feature Cube: Hierarchical Axiomatic Mapping. Groups correspond to the `:species` partition. **Laws**: *Identity* ($x \circ e = x$); *Absorber* ($x \circ z = z$); *Associative* ($(x \circ y) \circ z = x \circ (y \circ z)$); *Commutative* ($x \circ y = y \circ x$); *Idempotent* ($a \circ a = a$); *Inverse* ($a \circ a^{-1} = e$); *Monotonic* ($x < y \implies (x \circ z) < (y \circ z)$); *Logic* ($\chi : A \to \Omega$).

| **Species: bool** | Logical | | Bitwise | | | **Logic** |
|---|---|---|---|---|---|---|
| **Operator** | `\|\|` | `&&` | `\|` | `&` | `^` | $\Omega$ |
| **Identity** | `f` | `t` | `f` | `t` | `f` | |
| **Absorber** | `t` | `f` | `t` | `f` | — | |
| **Associative** | ✓ | ✓ | ✓ | ✓ | ✓ | |
| **Commutative** | ✓ | ✓ | ✓ | ✓ | ✓ | Classical |
| **Idempotent** | ✓ | ✓ | ✓ | ✓ | — | ($\Omega_{\mathbb{B}}$) |
| **Inverse** | — | — | — | — | ✓ | |
| **Monotonic** | ✓ | ✓ | ✓ | ✓ | — | |

| **Species: uint** | Arithmetic | | | Order | | | Bitwise | | | **Logic** |
|---|---|---|---|---|---|---|---|---|---|---|
| **Operator** | `+` | `*` | `%` | `min` | `max` | `<=` | `\|` | `&` | `^` | $\Omega$ |
| **Identity** | `0` | `1` | — | `~0u` | `0` | — | `0` | `~0u` | `0` | |
| **Absorber** | — | `0` | **Hz**$^{D}$ | `0` | `~0u` | — | `~0u` | `0` | — | Classical |
| **Associative** | ✓$^{M}$ | ✓$^{M}$ | — | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ($\Omega_{\mathbb{B}}$) |
| **Commutative** | ✓ | ✓ | — | ✓ | ✓ | — | ✓ | ✓ | ✓ | |
| **Idempotent** | — | — | — | ✓ | ✓ | ✓ | ✓ | ✓ | — | |
| **Inverse** | ✓$^{M}$ | — | — | — | — | — | — | — | ✓ | |
| **Monotonic** | **Hz**$^{W}$ | **Hz**$^{W}$ | — | ✓ | ✓ | ✓ | — | — | — | |

| **Species: int** | Arithmetic | | | Order | | | Bitwise | | | **Logic** |
|---|---|---|---|---|---|---|---|---|---|---|
| **Operator** | `+` | `*` | `%` | `min` | `max` | `<=` | `\|` | `&` | `^` | $\Omega$ |
| **Identity** | `0` | `1` | — | `I_MAX` | `I_MIN` | — | `0` | `-1` | `0` | |
| **Absorber** | — | `0` | **Hz**$^{D}$ | `I_MIN` | `I_MAX` | — | `-1` | `0` | — | Classical |
| **Associative** | **Hz**$^{U}$ | ✓ | — | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ($\Omega_{\mathbb{B}}$) |
| **Commutative** | ✓ | ✓ | — | ✓ | ✓ | — | ✓ | ✓ | ✓ | |
| **Idempotent** | — | — | — | ✓ | ✓ | ✓ | ✓ | ✓ | — | |
| **Inverse** | **Hz**$^{A}$ | — | — | — | — | — | — | — | ✓ | |
| **Monotonic** | ✓$^{P}$ | ✓$^{P}$ | — | ✓ | ✓ | ✓ | — | — | — | |

| **Species: double** | Arithmetic | | Order | | | **Logic** |
|---|---|---|---|---|---|---|
| **Operator** | `+` | `*` | `min` | `max` | `<=` | $\Omega$ |
| **Identity** | `0.0` | `1.0` | `Inf` | `-Inf` | — | |
| **Absorber** | `Inf` | `NaN` | `-Inf` | `Inf` | — | |
| **Associative** | $*$ | $*$ | ✓ | ✓ | ✓ | Ternary |
| **Commutative** | ✓ | ✓ | ✓ | ✓ | — | ($\Omega_{K3}$) |
| **Idempotent** | — | — | ✓ | ✓ | ✓ | |
| **Inverse** | ✓ | ✓ | — | — | — | |
| **Monotonic** | ✓ | ✓ | ✓ | ✓ | ✓ | |

$^{M}$ Modular success. $^{A}$ Asymmetry hazard (`INT_MIN`). $^{W}$ Wrap-around hazard.
$^{U}$ Undefined behavior (overflow). $^{D}$ Division hazard. $^{P}$ Partial success.
$*$*Fails bit-perfect associativity due to rounding/NaN.*

```cpp
// Reification of a raw lambda into an 'Atomic Arrow'
auto logic_gate = [](int x) -> bool { return x > 0; };

static_assert(IsArrow<decltype(logic_gate), int>,
              "Morphism Validated: The Atom of Composability is discovered.");
```

Listing 3.2: The Atom of Composability: Morphic Reification.

### 3.2.2 The `:morphism` partition: The Skeletal Arrow

Following the mapping established in Table 2.1, the `:morphism` partition reifies C++ callables as formal arrows $f : A \to B$. The core mechanism of this partition is **Signature Discovery**, which utilizes `std::invoke_result_t` and variadic template metaprogramming to extract domain and codomain metadata directly from the type system [5, 18].

```cpp
template <typename F, typename... Args>
concept IsArrow = requires {
    typename std::invoke_result_t<F, Args...>; // Codomain exists
    requires sizeof...(Args) > 0;              // Domain is non-empty
};
```

Listing 3.3: The Morphic Reification Hook (Compressed).

This "Atomic Arrow" serves as the fundamental unit from which all higher-level categorical structures are synthesized. By elevating the function object from an opaque procedural block to a structural entity, the library enables the compiler to reason about the composition and validity of transformations before any machine instructions are emitted.

#### The `:mereology` partition: The Language of Parts

The `:mereology` partition introduces the formal signature of parthood as the foundational structural primitive of the library. Following the mereological grounding of Stanisław Leśniewski, we treat parthood not as naive set membership, but as a primitive binary relation ($\leq$) that governs the structural identity of an entity. **If the Morphism is the atom of composability, then Mereology is the atom of hierarchy**, providing the skeletal vocabulary required for any downstream category to reason about sub-structures.

This build order is intentional and literature-aligned: structural mereology and order theory take part-whole or preorder structure as primitive, while logical calculi classify statements *about* those structures [22]. In categorical logic, internal logic is developed from already-established categorical structure (e.g., subobjects and classifiers), not the other way around [23, 21]. Therefore, in `dedekind`, `:mereology` provides the pre-logical relational floor that `:logic` later enriches with explicit truth-value objects.

By utilizing the signature discovery established in the previous section, the library reified parthood as a **Morphic Constraint**. A relation $R : A \times A \to \Omega$ is verified as a valid mereological inhabitant only if it satisfies the axioms of a **Partial Order**. As shown in Listing 3.4, the `IsPartRelation` concept acts as a formal verification gate, enforcing reflexivity ($a \leq a$), transitivity ($a \leq b \wedge b \leq c \implies a \leq c$), and antisymmetry ($a \leq b \wedge b \leq a \implies a = b$).

In implementation, the authoritative default relation is ordinary order (`std::less_equal<>`) with classical truth values; richer logical codomains are introduced downstream. This preserves a textbook-friendly dependency direction: define the relation first, then choose the logic used to classify its outcomes. The `:logic` partition (3.2.2) then addresses non-classical cases such as IEEE 754 singularities (e.g., `NaN <= NaN`) without inverting the foundational order [24, 13].

```cpp
/**
 * @brief The Mereological Signature: An abstract Partial Order.
```

```
 * @tparam R The Relation f: A x A -> Omega (Subobject Classifier)
 */
template <typename T, typename Op = std::less_equal<>, typename Omega = bool>
concept IsPartRelation = IsArrow<Op, T, T, Omega> &&
    requires(Op rel, T a, T b, T c) {
    // 1. Reflexivity: Pars partis est pars totius.
    static_assert(rel(a, a), "Reflexivity failed.");

    // 2. Transitivity: Hierarchical propagation.
    { rel(a, b) && rel(b, c) } -> std::convertible_to<bool>;

    // 3. Antisymmetry: Identity of indiscernible parts.
    { rel(a, b) && rel(b, a) } -> std::same_as<bool>;
  };

// SANITY CHECK: Integers satisfy the Mereological Signature.
static_assert(IsPartRelation<int, std::less_equal<>>,
              "Verified: Integrals form a valid Mereological hierarchy.");

// WARNING: Floats fail the Atomic Floor due to NaN reflexivity holes.
// static_assert(IsPartRelation<double, std::less_equal<>>); // Fails!
```

Listing 3.4: The Language of Parts: Verifying the Mereological Order.

### The `:logic` partition: The Subobject Classifier ($\Omega$)

With the dependency `:mereology` $\rightarrow$ `:logic`, logical species are introduced as an enrichment layer over a prior relational substrate. This follows the standard categorical story: once subobjects and order-like structure are in place, one can internalize truth as a classifier object and reason algebraically about propositions [23, 21, 24].

Before a "Body" (Set) or a "Path" (Sequence) can be reified, the `dedekind` library establishes the "Rules of Presence" through the **Subobject Classifier** $\Omega$. Following the algebraic logic of Rasiowa [24], we treat every logical system as a specific type of abstract algebra where truth-values act as the objects of a **Topos**. By reifying these systems via the `IsLogicalSpecies` concept, we prevent the "leaky abstractions" inherent in machine-level `bool` types—such as implicit integral promotion—while allowing for pluggable logical universes.

In the `ClassicalLogic` species, $\Omega$ is the binary prime $\{True, False\}$ (see Table 3.2), corresponding to the standard Boolean Topos where the **Law of Excluded Middle** (LEM) holds. However, to honestly model computational undecidability or the "truth-holes" inherent in floating-point boundaries, we introduce the `TernaryLogic` species. Based on Kleene's strong logic of indeterminacy ($K3$) [21], this logic introduces an **Unknown** state (0). Within `dedekind`, these logics are formalized as **Rigs**; the *Join* ($\vee$) serves as additive composition (Supremum), while the *Meet* ($\wedge$) serves as multiplicative composition (Infimum).

The necessity of `TernaryLogic` is most evident when observing the breakdown of the LEM in floating-point arithmetic. Following IEEE 754 [13], the `NaN` (Not-a-Number) state represents a singularity where standard relational morphisms fail to provide a binary partition. For a predicate $P(x) \coloneqq (x \leq 1.0)$ where $x = \text{NaN}$, classical C++ evaluates both $P(x)$ and its logical complement $\neg P(x)$ as `false`. This results in the paradox $P \vee \neg P = \bot$, a violation of classical tautology that `dedekind` resolves by elevating the subobject classification to $\Omega_{K3}$.

**The Lipschitz Boundary.** For signed integral types, the mapping from the infinite ring of integers to finite machine representations is only stable within a defined "Lipschitz domain" $[-2^{N-1}, 2^{N-1} - 1]$. By utilizing compiler built-ins (e.g., `__builtin_add_overflow`), the

Table 3.2: Truth Tables for Classical ($\Omega_{\mathbb{B}}$) and Kleene ($\Omega_{K3}$) Toposes.

Table 3.3: * Classical ($\Omega = \{0, 1\}$)

| $A$ | $B$ | $A \wedge B$ | $A \vee B$ | $\neg A$ |
|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 1 |
| 0 | 1 | 0 | 1 | 1 |
| 1 | 0 | 0 | 1 | 0 |
| 1 | 1 | 1 | 1 | 0 |

Table 3.4: * Kleene K3 ($\Omega = \{-1, 0, 1\}$)

| $A$ | $B$ | $A \wedge B$ | $A \vee B$ | $\neg A$ |
|---|---|---|---|---|
| -1 | -1 | -1 | -1 | 1 |
| -1 | 0 | -1 | 0 | 1 |
| -1 | 1 | -1 | 1 | 1 |
| 0 | 0 | 0 | 0 | 0 |
| 0 | 1 | 0 | 1 | 0 |
| 1 | 0 | 0 | 1 | -1 |
| 1 | 1 | 1 | 1 | -1 |

`SubobjectClassifier` detects boundary breaches and maps them to the **Unknown** state, ensuring that downstream transformations treat these singularities as formal subobject exclusions rather than undefined behavior.

```
// In the Dedekind Ternary Topos (Omega):
Ternary P      = Classifier<double>::eval(x <= 1.0); // Unknown
Ternary NOT_P  = !P;                                 // Unknown
Ternary LEM    = P || NOT_P;                         // Unknown

/** @brief Lipschitz Boundary Check for Signed Integers */
template <typename Op, typename T>
static constexpr Omega evaluate_arithmetic(T a, T b, Op op) {
    T result;
    if (__builtin_add_overflow(a, b, &result)) {
        return L::Unknown; // Lipschitz boundary breached
    }
    return L::True;
}
```

Listing 3.5: The NaN Truth-Hole and Lipschitz Boundary.

### Posetal Categories (`:posetal`)

The `:posetal` partition matures the abstract parthood defined in `:mereology` into a formal **Posetal Category**. It is critical to note that while traditional mathematics defines a Poset over a *set*, the `dedekind` build graph establishes these relations at Level 0—before the `:etcs` partition (Section 2.3.1) provides the set-theoretic axioms. Consequently, we treat posetal structures not as collections of elements, but as **Skeletal Categories** where, for any two objects $A$ and $B$, there exists *at most one* morphism $f : A \to B$.

In this skeletal framework, the existence of a unique arrow is governed by the parthood relation ($\leq$) as evaluated by the chosen Subobject Classifier $\Omega$. By anchoring these "Skeletal Orders" early, we provide the order-theoretic foundation required for the Lattices in Level 1 and the Dedekind-completeness proofs in Level 4. The `IsPosetal` concept (Listing 3.6) verifies that the relation is decidable within the species' specific logic, transforming a raw type into a structured coordinate system.

Furthermore, within this partition, semilattice and lattice concepts employ the **Two-Tier Architecture** (described below, Section 2.2.5) to separate structural verification from algebraic law certification. While `IsOrderMeetSemilattice<T>` checks only the syntactic interface (presence of a `meet` operation), the certified variant `IsCertifiedOrderMeetSemilattice<T>` adds trait-backed witnesses for commutativity, associativity, and idempotence, with final validation happening in the test suite.

```cpp
/**
 * @brief The Posetal Concept: A Skeletal Category.
 * @tparam T The Species (the objects of the category).
 * @tparam Rel The Partial Order (the unique morphisms).
 */
template <typename T, typename Rel>
concept IsPosetal = IsPartRelation<T, Rel, typename Classifier<T>::type> &&
  requires(Rel rel, T a, T b) {
    // The relation must return a value from the Species' Topos Omega
    { rel(a, b) } -> std::same_as<typename Classifier<T>::type>;
  };

// Verification: Standard integers under Classical Logic form a Posetal structure.
static_assert(IsPosetal<T, std::less_equal<int>>,
              "Verified: Integrals form a valid Posetal Category.");

// Structural Pruning: The compiler resolves categorical transitivity.
template <typename T, typename Rel>
requires IsPosetal<T, Rel>
constexpr auto check_chain(T a, T b, T c) {
    // If the arrows A->B and B->C exist...
    if constexpr (Rel{}(a, b) && Rel{}(b, c)) {
        // ...the arrow A->C is a categorical necessity.
        static_assert(Rel{}(a, c), "Categorical Transitivity Violation!");
        return true;
    }
    return false;
}
```

Listing 3.6: The Posetal Category: Orders as Unique Morphisms.

**Two-Tier Concept Architecture: Signature Verification vs. Law Certification.** Following the refactoring in PR #111, the `:posetal` partition adopts a **Two-Tier Concept Design** to address a fundamental tension in compile-time verification: while signature-level constraints (e.g., "does this type support the join operator?") are fully decidable via C++23 Concepts, algebraic law verification (e.g., "is join associative?") is undecidable for arbitrary types in the general case.

This architectural pattern partitions semilattice and lattice concepts into two complementary layers:

1. **Structural Concepts** (Signature-Level): These verify only the syntactic interface—e.g., that a type `T` supports binary operations `meet(a, b)` and `join(a, b)` with appropriate codomain types. Examples:

   - `IsOrderMeetSemilattice<T>` : Checks that `meet` exists and returns a `T`.
   - `IsOrderLatticeOperations<T>` : Checks composition of join/meet with absorption shape.

   These concepts generate zero runtime overhead and are suitable for generic programming where algorithmic correctness follows from the signature alone.

2. **Certified Variants** (Trait-Backed Laws): These extend structural concepts by requiring explicit trait-based legal *witnesses*. For instance, `IsCertifiedOrderMeetSemilattice<T, Meet>` requires:

   - The base `IsOrderMeetSemilattice` structural interface to hold.

- Trait witness `IsCommutative<T, Meet>::value` to be true (verifying at compile-time that the operation is marked as commutative in the `:species` registry).
- Trait witness `IsMereologicalMeetBand<T, Meet>` (verifying associativity and idempotence via the trait system).

This design shifts algebraic property verification away from undecidable `requires`-clauses toward explicit, composable trait witnesses that can be specialized by users for their custom types.

The key insight is that **undecidable equational properties (e.g., $a \wedge (b \vee c) = (a \wedge b) \vee (a \wedge c)$) are validated via the test suite as the canonical statement of law**, not as compile-time concept constraints. This follows the principle of *Semantic Honesty*: the library documents and validates which laws hold for which species, and users (or the CI pipeline) are responsible for verifying these properties at runtime on their concrete instantiations.

```
// Tier 1: Structural Interface (Signature-Only)
export template <typename T, typename Meet = decltype(std::ranges::min)>
concept IsOrderMeetSemilattice = requires(Meet meet, T a, T b) {
  { meet(a, b) } -> std::convertible_to<T>;
};

// Tier 2: Certified with Trait Witnesses
export template <typename T, typename Meet = decltype(std::ranges::min)>
concept IsCertifiedOrderMeetSemilattice =
    IsMereologicalMeetBand<T, Meet> &&   // Witnesses: associativity, idempotence
    IsCommutative<T, Meet>;               // Witness: commutativity trait

// Static verification using both tiers for int with std::ranges::min
static_assert(IsOrderMeetSemilattice<int, decltype(std::ranges::min)>);
static_assert(IsCertifiedOrderMeetSemilattice<int, decltype(std::ranges::min)>);

// Runtime validation happens in dedicated test suites (e.g., posetal_test.cpp)
TEST_CASE("Posetal: Semilattice Laws") {
  const int a = 7, b = 3, c = 9;
  const auto meet = std::ranges::min;

  // Commutativity
  CHECK(meet(a, b) == meet(b, a));

  // Associativity
  CHECK(meet(meet(a, b), c) == meet(a, meet(b, c)));

  // Idempotence
  CHECK(meet(a, a) == a);
}
```

Listing 3.7: Two-Tier Pattern: Structural vs. Certified Semilattice Concepts.

This pattern is now applied uniformly across all algebraic law verification in the library, providing both the clarity of compile-time type safety and the practical flexibility of runtime law validation.

### Small Categories (`:small`)

**Single-Species Category:** A single-species category is defined as a small category $\mathcal{C}$ containing exactly one object (the species) corresponding to exactly one C++ type or template parameter `T`. In this restricted setting, all morphisms $f : A \to A$ are endomorphisms, and the category inherits

the structure of a monoid where the binary operation is defined by morphism composition. This model provides a foundation for isolating species-specific logic before elevating interactions to functors, Cartesian products or coproducts, which is how mixed-type behaviour is modelled.

The `:small` partition formalises the transition from relational parthood to the general **Small Category**. While a *Posetal Category* (Section 2.2.5) restricts the morphism count between objects to at most one, the `IsSmallCategory` concept removes this constraint, reifying the axioms of **Categorical Composition** and **Identity**. In the `dedekind` ontology, a small category is treated as a finite registry of types and arrows that are visible to the compiler's proof-search engine.

By anchoring these axioms in Level 0, we provide the skeletal structure required for the *Functors* in Level 2. Crucially, the library enforces that for every object $X$ in the category, there exists an identity morphism $id_X$ and that the composition of arrows $f : A \to B$ and $g : B \to C$ is associative. As shown in Listing 3.8, this allows the compiler to verify that a transformation pipeline is not merely a sequence of function calls, but a mathematically valid path through the species' graph.

```
/**
 * @brief The Small Category Concept: Composition and Identity laws.
 * @tparam Cat The Category registry.
 */
template <typename Cat>
concept IsSmallCategory = requires {
    // 1. Identity: Every object X must have an identity arrow id_X.
    requires requires(typename Cat::Object X) {
        { Cat::identity(X) } -> IsArrow<typename Cat::Object, typename Cat::Object>;
    };

    // 2. Composition: f: A -> B and g: B -> C must compose to h: A -> C.
    requires requires(typename Cat::Arrow f, typename Cat::Arrow g) {
        requires std::same_as<typename SpeciesTraits<f>::Codomain,
                              typename SpeciesTraits<g>::Domain>;
        { Cat::compose(g, f) } -> IsArrow;
    };
};

// SANITY CHECK: The 'Atomic' category of C++ types satisfies Smallness.
static_assert(IsSmallCategory<CPlusPlusTypeCategory>);
```

Listing 3.8: The Small Category: Composition and Identity.

### Discrete Categories (`:discrete`)

The `:discrete` partition provides the degenerate categorical baseline where objects have identity arrows only. This is a pragmatic bridge between raw species registration and full morphic composition: it allows the compiler to treat a family of machine types as a category without forcing non-trivial arrows between them.

In implementation terms, this partition is low-ceremony but high-value: it supplies a canonical "zero-interaction" category used by tests and by later partitions when constructing examples that must preserve object identity without introducing additional algebraic structure.

### Universal Products and Coproducts (`:cartesian`)

To support multi-variable transformations and formal choice, the `:cartesian` partition reifies `std::pair` and `std::variant` through their universal properties. In the `dedekind` topos, a **Product** $(A \times B)$ is not merely a passive data container, but a structural requirement for the

existence of unique projections $\pi_1$ and $\pi_2$. By mapping the categorical product to `std::pair`, we establish the formal basis for binary algebraic operations—modeled as morphisms of the form $X \times X \to X$—ensuring that the "Rules of Both" are verified as categorical invariants.

Dually, the **Coproduct** $(A + B)$ is reified through `std::variant`, representing the "Rules of Choice." This allows for the categorical modeling of branching logic: a morphism out of a coproduct $f : A + B \to C$ is uniquely determined by a pair of morphisms $(f_A : A \to C, f_B : B \to C)$ via the mediator. By anchoring these universal constructions in the standard library's vocabulary, the library enables the Level 1 `:total` layer to define complex signatures while maintaining the zero-overhead performance of C++ machine types.

```
/**
 * @brief Categorical Product Verification:
 * Ensures T satisfies universal projection properties for A and B.
 */
template <typename T, typename A, typename B>
concept IsProduct = requires(T product) {
    { project<0>(product) } -> std::same_as<A>;
    { project<1>(product) } -> std::same_as<B>;
};

// Verification of the Categorical Product (std::pair)
static_assert(IsProduct<std::pair<int, int>, int, int>,
              "Verified: std::pair is a valid Categorical Product.");

/**
 * @brief Categorical Coproduct Verification:
 * Ensures T satisfies the Rule of Choice (Injectors).
 */
template <typename T, typename A, typename B>
concept IsCoproduct = requires(A a, B b) {
    { inject<A>(a) } -> std::same_as<T>;
    { inject<B>(b) } -> std::same_as<T>;
};

// Verification of the Categorical Coproduct (std::variant)
static_assert(IsCoproduct<std::variant<int, bool>, int, bool>,
              "Verified: std::variant is a valid Categorical Coproduct.");

// Binary Morphism Signature: X * X -> X
using Addition = Morphism<std::pair<int, int>, int, std::plus<int>>;
static_assert(IsArrow<Addition, std::pair<int, int>, int>);
```

Listing 3.9: The Universal Bridge: Mapping Products and Coproducts to C++ Primitives.

### Boundary Objects (`:limit`)

The `:limit` partition defines the logical boundaries of the category: the **Initial** (0) and **Terminal** (1) objects. Following the structuralist approach advocated by Lawvere [19], these are defined not by their internal contents, but by the existence of unique morphisms to or from these boundaries. In the C++23 context, we reify the Terminal object as `std::monostate` (or a unit type), representing a state of pure existence without information.

This abstraction allows for the formal definition of "elements" as morphisms from the terminal object $(1 \to X)$. In `dedekind`, a constant value is not a static literal, but a *global element*—a mapping that selects a specific inhabitant of a species. Dually, the Initial object (0) represents the absolute annihilator; a morphism from the initial object $0 \to X$ is the categorical equivalent of an unreachable code path. By anchoring these boundaries early in the build graph, we provide

the formal basis for the :etcs partition to reason about set membership as a structural relation.

```
/** @brief The Terminal Object (1): Pure existence. */
using One = std::monostate;
static_assert(IsTerminalObject<One>);

/** @brief Defining a "Global Element" as a Morphism 1 -> X */
auto five = Morphism<One, int>([](One) { return 5; });
static_assert(IsArrow<decltype(five), One, int>);

/** @brief Initial Object (0): The unreachable annihilator. */
using Zero = std::nullptr_t; // Categorical void / Unreachable
static_assert(IsInitialObject<Zero>);

// Verification: Every species T has a unique morphism T -> 1 (Terminality)
static_assert(HasUniqueMorphismTo<int, One>);
```

Listing 3.10: The Boundary Objects: Reifying 0 and 1 in C++23.

#### Universal Constructions and Pullbacks (:pullback)

To support relational reasoning and intersections, the :pullback partition introduces the universal property of the **Pullback**. As noted in Pierce [25], pullbacks are the categorical tool for handling constraints and fiber products. By reifying the pullback, the library enables the formal definition of **Equalizers**, which are essential for identifying the rounding-error singularities in the :numeric partition.

In the dedekind ontology, the pullback $P$ of two morphisms $f : A \to C$ and $g : B \to C$ represents the subset of the product $A \times B$ where the morphisms coincide ($f(a) = g(b)$). This construction is the categorical engine for set intersection; by hoisting this requirement into the type-system, we allow the compiler to perform **Symbolic Equalization**, identifying when two computational paths are structurally identical or divergent before any floating-point operation is executed.

```
/** @brief Pullback P of f: A -> C and g: B -> C */
template <typename f, typename g>
using Pullback = Set<Product<Domain<f>, Domain<g>>, [](auto pair) {
    return f{}(project<0>(pair)) == g{}(project<1>(pair));
}>;

// An Equalizer is a pullback of f and g over the same codomain
using Eq = Equalizer<f, g>;
static_assert(IsLimit<Eq>);
```

Listing 3.11: Pullbacks and Equalizers: The Engine of Constraint.

### 3.2.3 Level 1: Space and Action

The transition from Level 0 to Level 1 represents the "Maturation of Species." Following the Polish School of Logic as articulated by Ajdukiewicz [26], we treat mathematical structures as active apparatuses—or conceptual orientations—that determine the image of the computational world. This aligns with the restriction category approach [27], where totality is not a default state but a property verified against a specific subobject support.

#### The :total partition: The Property of Totality

The :total partition formalizes *Morphic Totality*. In our ontology, a morphism $f : A \to B$ is verified as a TotalMorphism only if it is defined across its entire domain without triggering

a subobject exclusion (e.g., overflow or division by zero). Following the restriction category framework [27], we treat totality as a property verified against a specific subobject support, ensuring that the "active apparatus" of the morphism is physically stable within the machine's constraints.

```
/** @brief Formal verification that a morphism is defined for the entire species. */
export template <typename F, typename A, typename B>
concept IsTotalArrow = IsArrow<F, A, B> && requires(F f, A x) {
  // Totality check: The SubobjectClassifier must return 'True'
  requires std::same_as<typename SubobjectClassifier<A>::Omega, bool>;
  { SubobjectClassifier<A>::evaluate_total(x) } -> std::same_as<bool>;
};

export template <IsSpecies A, IsSpecies B, typename Func>
struct TotalMorphism : Morphism<A, B, Func> {
  static_assert(IsTotalArrow<Func, A, B>, "Totality Error: Function is not total.");
};
```

Listing 3.12: Totality Verification and the Total Morphism wrapper.

### Total Algebraic Layer (within `:total`): The Laws of Harmony

Building upon the total mappings established in Level 1a, the algebraic layer within `:total` materializes the "Pure Ideals" of abstract algebra. In this partition, species "mature" as they gain axioms, moving from the primal **Magma** to the perfect symmetry of an **Abelian Group**. A key architectural decision in `dedekind` is the decoupling of *associativity* from *invertibility*: while a **Monoid** provides an identity to an associative Semigroup, a **Unital Magma** (or **Loop** if invertible) provides an identity even when associativity is rejected [28]. A **Group** is thus reified as the perfect synthesis of these two paths, as shown in Table 3.5.

This leads to the **Honest Rejection** policy: while `unsigned int` addition is verified as a total Abelian Group $(\mathbb{Z}/2^n\mathbb{Z}, +)$, signed `int` is relegated to a **Unital Magma**. The library discovers the identity element (0) as a safe invariant, yet rejects the **Associative** and **Distributive** axioms. Because signed overflow constitutes undefined behavior, the C++ abstract machine cannot guarantee structural harmony under re-ordering; the library correctly identifies such operations as **Hazards** [29], preserving the integrity of the total domain.

Table 3.5: The Path to Symmetry: Algebraic Species in `dedekind.category`. The **Group** represents the synthesis of the associative path (Monoid) and the invertible path (Loop).

| Concept | Structure | Axiomatic Requirements | C++ Concept |
|---|---|---|---|
| **Magma** | $\langle T, \circ \rangle$ | Closure: $\forall a, b \in T : a \circ b \in T$ | `IsMagma` |
| **Semigroup** | $\langle T, \circ \rangle$ | Magma + Associativity | `IsSemigroup` |
| **Monoid** | $\langle T, \circ, e \rangle$ | Semigroup + Identity | `IsMonoid` |
| **Loop** | $\langle T, \circ, e \rangle$ | Magma + Identity + Unique Division | `IsLoop` |
| **Group** | $\langle T, \circ, e \rangle$ | Monoid + Loop | `IsGroup` |
| **Abelian Group** | $\langle T, \circ, e \rangle$ | Group + Commutativity | `IsAbelianGroup` |

```
/** @section Truth_Anchors */

// 1. Verification of the Modular Ideal (Z/2^nZ, +)
static_assert(IsAbelianGroup<unsigned int, std::plus<>>,
              "Axiom Error: Unsigned addition is a perfect Abelian Group.");

// 2. The Honest Rejection (Hazardous Domain)
// Signed addition provides an identity (Unital Magma),
```

```
// but is rejected as a Semigroup due to the UB overflow hazard.
static_assert(has_identity_v<int, std::plus<>>,
              "Maturation: Signed identity (0) is safe and discoverable.");

static_assert(!is_associative_v<int, std::plus<>>,
              "Honesty Check: Signed associativity rejected due to UB.");

static_assert(!IsSemigroup<int, std::plus<>>,
              "Taxonomy: Signed int addition is not a Semigroup.");

// 3. The Boolean Ring (Characteristic 2)
static_assert(characteristic_v<bool> == 2,
              "Axiom Error: Boolean Ring must have characteristic 2.");

static_assert(is_idempotent_v<bool, std::logical_or<>>,
              "Axiom Error: Boolean OR must be idempotent.");

// 4. Distributive Handshake
static_assert(is_distributive_v<unsigned int, std::multiplies<>, std::plus<>>,
              "Axiom Error: Unsigned * must distribute over +.");

static_assert(!is_distributive_v<int, std::multiplies<>, std::plus<>>,
              "Honesty Check: Distributivity rejected for signed hazards.");
```

Listing 3.13: Structural Maturation and the Honest Rejection in dedekind.category.

### Monoid Actions and Modules (`:action`)

The `:action` partition materializes the concept of a Species being acted upon by an Algebraic Structure. This partition defines the `IsSemiModule` and `IsModule` concepts, where a **Monoid** (from the `:total` algebraic layer) defines a transformation rule over a target Species.

Following the pedagogical tradition of Stammbach [30], we treat linear algebra not as a series of matrix operations, but as the study of structured actions. In this ontology, we distinguish between **Internal Composition** (operations within a species) and **External Influence** (actions upon a species). This "coordinate-free" approach [31] is critical for transitioning from pure algebraic laws to the modeling of transformations across spaces, establishing the formal basis for the linear transformations and tensor structures later introduced in the `:geometry` module.

### The `:functor` partition: Mappings between Categories

**Functors and Endofunctors:** A functor $F : \mathcal{C} \to \mathcal{D}$ is modeled as a morphism between categories. In our implementation, $F$ acts as an "arrow between species" mapping arrows from $\Sigma_{cat}$ to $\mathrm{T}_{cat}$. Crucially, we distinguish **endofunctors** where $\Sigma_{cat} = \mathrm{T}_{cat}$. In this library, all Monads and Comonads are strictly required to be endofunctors to ensure the coherence of multiplication ($\mu : T^2 \Rightarrow T$) and comultiplication ($\delta : W \Rightarrow W^2$).

Following the fractal progression of the library, the `:functor` partition reifies the algebraic laws of Level 1 to act upon the categories themselves. In `dedekind`, a Functor is treated not as a container, but as a structural morphism $F : \mathcal{C} \to \mathcal{D}$ that preserves the composition of arrows and the identity of objects. By elevating the notion of mapping to a category-level operation, we allow the library to reason about transformations across different ontological domains.

The implementation centers on a high-level `fmap` dispatcher. Leveraging C++23 template-template parameters (e.g., `template <typename> typename F`), the dispatcher ensures that functorial mapping is a zero-overhead operation. This architecture shields the user from the underlying machine representation while maintaining strict categorical rigor. By verifying that

$F(g \circ f) = F(g) \circ F(f)$ at compile-time, the compiler ensures that the structural integrity of the mathematical model remains invariant under transformation.

```
/**
 * @brief The Functorial Dispatcher (fmap).
 * Maps an Arrow (f: A -> B) to a lifted Arrow (F(f): F<A> -> F<B>).
 */
export template <template <typename> typename F, IsArrow Arrow>
constexpr auto fmap(Arrow f) {
    return lifted_morphism<F>(f);
}

// Verification: Functors must preserve the Identity Morphism.
// Here, 'F' represents a generic skeletal functor.
static_assert(PreservesIdentity<F, int>);
```

Listing 3.14: Functorial Mapping: Mappings as zero-overhead dispatchers.

### The `:natural` partition: Natural Transformations

**Natural Transformations:** Natural transformations are treated as 2-cells, acting as morphisms between functors. For any two functors $F, G : \mathcal{C} \to \mathcal{D}$, a natural transformation $\alpha : F \Rightarrow G$ provides a family of morphisms in the target category $\mathcal{D}$ such that for every arrow $f : A \to B$ in $\mathcal{C}$, the naturality square commutes: $\alpha_B \circ F(f) = G(f) \circ \alpha_A$.

To complete the Level 2 bridge, the `:natural` partition introduces **Natural Transformations**—morphisms between functors. This provides the mathematical basis for "component-wise" transformations that remain invariant under functorial mapping, satisfying the naturality square: $\alpha_Y \circ F(f) = G(f) \circ \alpha_X$. By formalizing these mappings, the library ensures that structural changes—such as the transition from a skeletal species to an effectful one—are consistent across all objects in the category.

The `:natural` partition is a strict technical prerequisite for the `:monad` partition, as it establishes the formal requirements for the unit ($\eta$) and multiplication ($\mu$) transformations. In `dedekind`, naturality is not merely a convention but a verified invariant. Leveraging C++23 Concepts, we implement the `IsNaturalTransformation` constraint to verify that a structural mapping $\alpha$ commutes with the `fmap` of its source and target functors. This verification ensures that the underlying structural integrity is maintained across different functorial contexts without "species-leak" or logical drift.

```
/**
 * @concept IsNaturalTransformation
 * @brief Formal verification of the naturality square.
 */
export template <typename Alpha, template <typename> typename F,
                 template <typename> typename G>
concept IsNaturalTransformation = requires(Alpha alpha) {
    // For any morphism f: A -> B, the following must commute:
    // alpha<B> o fmap<F>(f) == fmap<G>(f) o alpha<A>
    requires CommutesWithFmap<Alpha, F, G>;
};

// Skeletal unit transformation (eta): Id -> F
export template <template <typename> typename F>
struct UnitTransformation;
```

Listing 3.15: The Naturality Square: Verifying consistency between Functors.

### The `:monad` partition: The Ω-Monad

**Monads and Comonads as Endomorphisms:** In this library, Monads and Comonads are defined strictly over endofunctors $T : \mathcal{C} \to \mathcal{C}$. This "endo" property is a foundational requirement for the coherence of the natural transformations: the monadic multiplication $\mu : T \circ T \Rightarrow T$ and the comonadic comultiplication $\delta : W \Rightarrow W \circ W$ are only well-defined when the source category $\Sigma_{cat}$ and target category $\mathrm{T}_{cat}$ are identical. Consequently, any type satisfying the `IsMonad` or `IsComonad` concept must provide a compile-time proof that its underlying functor maps the species category back to itself.

**Frobenius Symmetry:** A structure $F$ is identified as a Frobenius type when it simultaneously satisfies the requirements of a monad and a comonad in a way that the monoid $(\eta, \mu)$ and comonoid $(\varepsilon, \delta)$ structures are compatible. This symmetry is expressed through the Frobenius law:

$$\mu \circ (\delta \times Id) = \delta \circ \mu = \mu \circ (Id \times \delta)$$

In the context of the C++ type system, types such as finite coproducts (`std::variant`) or the List monad (`std::vector`) exhibit this symmetry, where the "wrapping" and "unwrapping" of context are adjoint operations that preserve the underlying categorical energy.

While Level 1 defines absolute laws, real-world computation requires a mechanism for handling "effectful" transformations. Following the constructions of Kleisli [20] and Moggi [6], we implement the **Kleisli Triple** $(\eta, \gg=)$ as the fundamental unit of composition. In the `:monad` partition, we unify the "Rules of Presence" from the `:logic` topos with this machinery to define the **Ω-Monad**.

As illustrated in Listing 3.16, we extract the `IsPotential` concept to create a structural isomorphism between standard C++ types and the `dedekind Partial<T>` container. To resolve the lack of native higher-kinded types (HKTs) in C++, we utilize an "Action-First" bootstrapping strategy: the `fmap` functorial mapping is derived as an *epi-phenomenon* of the underlying monadic "Push" $(m \gg= \eta \circ f)$. This ensures that any species providing a *bind* operator $(\gg=)$ automatically matures into a *Functor* through *Synthetic Inference*.

```
export template <typename R>
concept IsPotential = requires(R r) {
    typename R::value_type;
    { *r } -> std::convertible_to<typename R::value_type&>;
    { presence_of(r) } -> LogicalValue; // Bridge to Omega
};

// The Universal Highway: Works for any IsPotential result
export template <IsPotential M, typename F>
constexpr auto operator>>=(const M& m, F f) {
    using L = typename GetLogic<M>::type;
    if (presence_of(m) != L::True) return empty<M>(presence_of(m));
    return f(*m);
}
```

Listing 3.16: The Ω-Monad: Structural Parity and the Universal Highway.

### The `:kleisli` partition: The Universal Highway

The `:kleisli` partition introduces the formal machinery for *Highway composition* (operator »). Following the functional tradition of the "Fish Operator" (`>=>`) [32], `dedekind` adapts this to the C++23 context by leveraging the left-associativity of `operator»` to model the directional vector of computation across the ontology.

To resolve the lack of native higher-kinded types (HKTs) in C++, the library utilizes an *Action-First* bootstrapping strategy. By defining Monads as Extension Systems (Kleisli Triples

consisting of $\eta$ and $\gg=$), we utilize operator overloading on concrete objects to trigger Argument-Dependent Lookup (ADL) at the call site. This allows the Level 0 foundation to instantiate a derived `fmap` as an epi-phenomenon of the underlying monadic "Push" ($m \gg= \eta \circ f$).

The resulting *Highway Notation* reifies categorical data flows as directional vectors:

- `value » into<F>` : Represents $\eta$ (Unit), lifting a species into a context.
- `box « extract<F>` : Represents $\varepsilon$ (Counit), sampling a species from a context.
- `box »= f` : Represents Bind, the forward monadic composition.
- `box «= f` : Represents Extend, the contextual comonadic composition.

Because assignment-style operators in C++ are right-associative, the partition also provides fish-style aliases for value-level extension: `box » f` as an alias of bind and `box « f` as an alias of cobind. This supports readable left-associated pipelines (e.g., `box » f » g` and `box « f « g`) while preserving the same Kleisli and co-Kleisli semantics. A minimal implementation example is shown in Listing 3.17, with regression coverage in `src/test/cpp/modules/dedekind/category/kleisli_test.`

```
Box<int> value{5};
auto bound = value >> [](int x) { return Box<int>{x + 10}; };
auto chained = value >> [](int x) { return Box<int>{x * 2}; }
                     >> [](int x) { return Box<int>{x + 1}; };

Box<int> context{8};
auto cobound = context << [](Box<int> b) { return b.value - 2; };
auto cochained = context << [](Box<int> b) { return b.value / 2; }
                         << [](Box<int> b) { return b.value + 3; };
```

Listing 3.17: Value-level fish aliases for Kleisli and co-Kleisli extension.

Within the verified module graph, the LLVM backend utilizes this structural transparency to perform *Highway Pruning*. if a total function is composed with a partial operation statically known to return an `Unknown` or `NaN` state, the compiler identifies the unreachable path and optimizes the Highway into a branch-free instruction sequence.

### The `:partial` partition: The Pragmatic Rescue

The `:partial` partition performs the "Categorical Rescue" of species that failed the strict requirements of the total domain. This stage formalises the transition from the *Platonic* to the *Pragmatic*: acknowledging that while hardware-level operations may be "broken" (e.g., via overflow or division by zero), they are **consistently broken**. Following the formalisations of the *nLab* community [33], we treat these consistent failures not as exceptions, but as structured data.

In the current build graph, `:numeric` is imported before `:partial` to provide hardware-level boundary classifiers used by partial algebra; the exposition here preserves the conceptual narrative while the implementation follows dependency order.

Following topos-theoretic foundations [21], we treat a partial morphism $f : A \rightharpoonup B$ as a total mapping from a *subobject* $S \hookrightarrow A$. Here, $S$ represents the **Support** defined by the characteristic function $\chi : A \to \Omega_{K3}$ provided by the `:logic` partition. Within this restricted domain, we define *locally total* structures, such as **Partial Groups** or **Partial Monoids**. These structures guarantee that the "Laws of Harmony" hold strictly on the support $U = \{a \in A \mid \chi(a) = \top\}$, ensuring that even when navigating the messy reality of hardware, we remain within a rigorous algebraic landscape.

By reifying the `Partial<T>` "Rescue Pod" as an inhabitant of the $\Omega$-Monad, the library enables the composition of these partial morphisms via the universal Highway established in the

`:kleisli` partition. The bind operator ($\gg=$) acts as the monadic gatekeeper: it consumes the `Ternary` presence, ensuring that any "Truth-Hole" is propagated through the Kleisli chain. This transforms inconsistent machine behavior into a rigorous, composable algebraic flow.

```
// The "Rescue": int addition is recognized as a Partial Abelian Group.
static_assert(IsPartialAbelianGroup<int, std::plus<int>>);

// Composition in the Kleisli Category via the Highway:
auto f = partial_plus(1, 2);          // Result: 3 (True)
auto g = partial_plus(f, INT_MAX);    // Result: Unknown (Overflow)
auto h = partial_plus(g, 1);          // Result: Unknown (Propagation)
```

Listing 3.18: The Kleisli Lift: Reclaiming the Lipschitz Boundary.

### Partial Algebraic Layer (within `:partial`): The Generalised Reality

Following the monadic Kleisli lift, the algebraic layer within `:partial` generalizes the algebraic laws of Level 1 to handle the "broken" reality of hardware. In this layer, the strict requirements of absolute closure and identity are relaxed into **Partial Monoids** and **Partial Groups**. Unlike their total counterparts, these structures are only guaranteed to hold within the specific support $S \hookrightarrow A$ provided by the `:logic` and `:partial` partitions.

The core objective of this partition is to verify that a species is "consistently broken." By reifying partiality through the $\Omega$-Monad, `dedekind` ensures that if a result is defined (i.e., its presence is $\top$), it must still satisfy the required algebraic identities. For instance, a **Partial Monoid** must remain associative wherever its composition is defined. This allows the library's dispatcher to safely propagate `Unknown` or `NaN` states through the algebraic pipeline without triggering undefined behavior. By enforcing these laws within the support, we ensure that the "Laws of Harmony" are not discarded, but are instead localized, allowing the compiler to perform structural pruning even in the presence of indeterminate data.

```
// Verification: A Partial Monoid is associative on its support.
export template <typename T, typename Op>
concept IsPartialMonoid = IsPartialMagma<T, Op> &&
                          HasLocalIdentity<T, Op> &&
                          IsLocallyAssociative<T, Op>;

// Signed integers: Total Magma (REJECTED), Partial Abelian Group (SUCCESS).
static_assert(IsPartialAbelianGroup<int, std::plus<int>>);
```

Listing 3.19: Local Totality: Verifying the Laws of Harmony on the Support.

### The `:numeric` partition: The Material Landing

The `:numeric` partition represents the final "landing" of the abstract theory back onto the hardware substrate. While the `:partial` partition (including its internal algebraic layer) establishes the general laws for inconsistent structures, the `:numeric` partition provides the specific mapping for IEEE 754 floating-point types and bounded integers.

In this partition, the library utilizes the `Partial<T>` rescue pod to formally model the non-associative nature of `double` and the non-reflexive behavior of `NaN`. By anchoring these hardware-specific behaviors as first-class inhabitants of a partial monoid, we ensure that the "messy reality" of rounding errors and overflows is not an exception to the theory, but a verified and traceable component of the categorical flow. This is achieved through "Safety Certificates" for specific machine species, which allow the `dedekind` dispatcher to dynamically enable or disable algebraic optimizations (such as expression reordering) based on the material reality of the underlying type.

Finally, we encounter the curious case of *Numerical Species* (IEEE 754). These types are effectively categories where the subobject classifier $\Omega$ is internalized within the species itself via states such as `NaN` and `Inf`. While these structures appear to be total morphisms $\mu : \mathbb{R}_f \times \mathbb{R}_f \to \mathbb{R}_f$ in their machine signature, they are semantically partial. Specifically, they fail the **Equalizer** conditions required for bit-perfect associativity; the rounding error ensures that $(a + b) + c$ and $a + (b + c)$ are not generally isomorphic. This progression—from the absolute (*Total*), through the filtered (*Partial*), to the exceptional (*Numerical*)—allows us to build a rigorous ontology that respects both categorical rigor and the physical constraints of the C++ host language.

```
// IEEE 754 Addition: Partial Abelian Group (SUCCESS).
static_assert(IsPartialAbelianGroup<double, std::plus<double>>);

// However, it is NOT a Total Monoid due to rounding error.
static_assert(!IsTotalMonoid<double, std::plus<double>>);

// Safety Certificate: Disables reassociation for floating-point.
static_assert(!is_associative_v<double, std::plus<double>>);
```

Listing 3.20: The Numeric Landing: Verifying the IEEE 754 Partial Monoid.

### The `:etcs` partition: The Set-Theoretic Interface

The final stage of the foundational build anchors Lawvere's Elementary Theory of the Category of Sets (ETCS) [19]. This partition defines a set not by its internal elements, but by its universal properties, utilizing C++23 Concepts to enforce well-pointedness and the existence of power objects.

The core innovation of this partition is the reification of the *Set-Builder* notation. By representing a set $S = \{x \in X \mid P(x)\}$ as a `Stateless Constraint Type`, we transform the C++ translation unit into a symbolic laboratory.

```
// Define a predicate: x is even and greater than 10
auto P = [](auto x) { return (x % 2 == 0) && (x > 10); };

// Materialize a categorical set object over the ambient species int
const auto even\_greater\_ten = dedekind::category::ambient\_set<int>(P);

static\_assert(dedekind::category::IsSet<decltype(even\_greater\_ten)>);
static\_assert(even\_greater\_ten.\u03C7(12) == true);
```

Listing 3.21: Reifying the Set-Builder via Stateless Constraints.

By hoisting these mereological invariants into the type signature, we enable *Symbolic Equalization*. The compiler can decompose and optimize a set based on its logical essence rather than its storage layout, allowing contradictory set-builder predicates to collapse into zero-instruction machine code. This ensures that the performance of the "Naked Loop" is always mathematically justified by the prior formal certification of the ETCS axioms.

### Topoi and Internal Logic (`:topoi`)

The `:topoi` partition packages the categorical preconditions for internal reasoning: subobject classifiers, finite limits, and exponentials are treated as first-class structural witnesses. This partition is the conceptual hinge between pure category structure and logic-as-algebra in the implementation.

Operationally, `:topoi` complements `:etcs`: while ETCS gives the user-facing set-theoretic facade, the topos partition provides the internal categorical contracts that justify that facade. Keeping both partitions explicit in the build graph prevents silent drift between foundational definitions and set-level APIs.

## 3.3 Results

The `dedekind.category` module is verified in the CI pipeline through the `category_test` and `kleisli_test` suites. Key verified properties are summarised in Table 2.1 (primary formal correspondences), Table 3.1 (Feature Cube of algebraic species), and Table 3.5 (algebraic hierarchy). The module dependency graph is shown in Figure 3.2.1. All 15 partitions compile and link without errors, and coverage for the module is at or above the project baseline.

# Chapter 4

# dedekind.sets

## 4.1 Introduction

The `dedekind.sets` module implements the Elementary Theory of the Category of Sets (ETCS) [19, 34] through a structural, mereological approach. Set objects are defined by their membership predicates rather than by extensional storage, enabling the C++ compiler to reason symbolically about union, intersection, complement, and subset operations. The module occupies the second position in the build graph, immediately downstream of `dedekind.category`, and provides the set-theoretic substrate for all subsequent modules.

Key references are Lawvere and Rosebrugh [34], structural mereology [22], and Cartesian closed categories [35].

## 4.2 Implementation

**Implementation note (terminology and scope).** The current implementation separates `dedekind.category:mereology` from `dedekind.sets:mereology` on purpose. The category partition now exposes a minimal, shared core for parthood and lattice-like operations used by downstream modules, while the sets partition preserves richer compatibility constraints required by existing tests and clients.

In textbook order/lattice terminology, a join- or meet-semilattice is associative, commutative, and idempotent. Our present sets-side naming is therefore transitional: some operation-level concepts intentionally retain weaker constraints to avoid API churn during the ongoing ETCS-aligned refactoring. A dedicated taxonomy pass is planned to promote textbook names for commutative structures and reserve separate names for non-commutative idempotent operations.

For related typing and set-operation design directions, see: *Monadic Intersection Types, Relationally, and Ordered* (`https://dl.acm.org/doi/10.1145/3777483`) and references therein.

**Key implementation insights.** Three implementation choices proved decisive for the practical behavior of the `sets` module. First, we keep the set-builder predicate as an explicit structural parameter, so contradictory constraints can be simplified by the compiler before runtime materialization; this mirrors the ETCS idea that object identity is relational, not representational [19, 34]. Second, we preserve a strict layering from `category` to `sets` so that order/parthood witnesses are available as immutable build artifacts (BMIs), reducing circular meta-program instantiation pressure in large translation units. Third, we use two-tier certification in practice: structural concepts establish interface decidability at compile time, while semantic laws are validated in tests, following the "instrumented model validation" posture rather than claiming full theorem-prover completeness [15, 36].

For numerics-facing set operations, this layering is important because IEEE 754 singularities (`NaN`, overflow neighborhoods) are handled as classifier-level exceptions rather than hidden

runtime accidents [13, 37].

**Partial ETCS Alignment Strategy.** Following PR #111, the `sets` module implements a **Partial ETCS Alignment Strategy**, reifying 8 of the 10 Lawvere axioms of the Elementary Theory of the Category of Sets. This pragmatic approach acknowledges that full ETCS alignment at the type-level is achievable without sacrificing practical usability; the deferred axioms (Axiom 7: Unique Factorization through Pullbacks, and Axiom 10: Well-Pointedness) are strategically deferred to future levels of the library where higher-order categorical machinery becomes available.

The 8 active axioms are reified through explicit trait witnesses in the form of axiom composites:

1. **Axiom 1 (Composition)**: `HasAxiom1Composition<T>` verifies that the composition of characteristic functions is associative.
2. **Axiom 2 (Identity)**: `HasAxiom2Identity<T>` ensures that every set admits an identity morphism.
3. **Axiom 3 (Product Existence)**: `HasAxiom3ProductExistence<T>` certifies Cartesian products of sets.
4. **Axiom 4 (Disjoint Union)**: `HasAxiom4DisjointUnion<T>` verifies coproducts (disjoint unions).
5. **Axiom 5 (Equalizer)**: `HasAxiom5Equalizer<T>` ensures the existence of equalizer subobjects.
6. **Axiom 6 (Exponentiation)**: `HasAxiom6Exponentiation<T>` reifies the power object via the subobject classifier.
7. **Axiom 8 (Choice Function)**: `HasAxiom8ChoiceFunction<T>` provides explicit choice for nonempty fibers.
8. **Axiom 9 (Natural Numbers Object)**: `HasAxiom9NaturalNumbers<T>` ensures inductive structure on appropriate species.

The deferred Axioms 7 and 10 depend on pullback diagrams and the global section property, respectively, which require more sophisticated higher-order categorical formalism not yet integrated into the library's architecture. By explicitly documenting this partial alignment, the library provides a clear roadmap for future extensions while maintaining mathematical honesty about the current scope of ETCS formalization.

### 4.2.1 The `:boundaries` partition

The `:boundaries` partition defines the extremal set objects for a given species and logic: the empty set $\varnothing$ and the ambient/universal set $\Omega$. These boundary objects are the neutral and absorbing anchors for set operations, and they provide the concrete entry point for ETCS-style subobject reasoning in downstream partitions.

In practical terms, this partition stabilizes the API for identities used by union, intersection, and complement while keeping the carrier and classifier explicit at the type level.

### 4.2.2 The `:expressions` partition

The `:expressions` partition is the set-builder workhorse. It provides intensional sets as predicates, symbolic variable bindings, logical predicate composition, subset witnesses, Cartesian products, relations, set-functions, and power-set construction.

Recent ETCS-aligned extensions in this partition include point-wise subset evidence, relation witnesses, and function single-valuedness checks, all expressed as zero-overhead compositional constraints in the type system.

### 4.2.3 The `:mereology` partition

The `:mereology` partition in `dedekind.sets` specializes part-whole operations for concrete set expressions. While category-level mereology supplies shared foundations, this partition preserves richer set-specific compatibility constraints and operation-level laws required by client code and regression tests.

This split is intentional during the ongoing taxonomy alignment: it allows the project to converge toward textbook naming and law-strength classifications without destabilizing existing semantics.

### 4.2.4 The `:singleton` partition

The `:singleton` partition reifies one-element sets as first-class objects. Singletons are the minimal extensional witnesses used to bridge element-level statements with set-level morphisms, especially in tests that validate ETCS boundary behavior and comprehension syntax.

Because singleton construction is trivial but semantically central, this partition acts as a low-risk anchor for onboarding examples and for validating element-membership notation against the broader intensional framework.

### 4.2.5 The `:set` partition

To ground this theoretical approach, Table 1 maps the fundamental axioms of the Elementary Theory of the Category of Sets (ETCS) – as codified in the pedagogical treatment by Lawvere and Rosebrugh [34] – to their structural equivalents in `dedekind`. This mapping ensures that the library is not merely a numerical simulation, but a direct implementation of algebraic laws within the host language's type system. In a sense, `dedekind` hopes to be a *compiler* for mathematical concepts.

Table 4.1: Complete Mapping of ETCS Axioms to C++23 Structural Invariants

| **ETCS Axiom** | **C++ Language Feature** | **Structural Invariant** |
|---|---|---|
| 1. Composition | `std::invoke` / `operator»` | `IsSemigroupoid<T, Op>` |
| 2. Identity | `identity_trait<T, Op>` | `IsSmallCategory<T, Op>` |
| 3. Terminal Object | `s(x) -> Logic::top` | `IsTerminalObject<T>` |
| 4. Well-Pointedness | `operator()` (Element access) | `HasAxiom4WellPointedness` |
| 5. Cartesian Product | `std::tuple` / `std::pair` | `HasProduct<A, B>` |
| 6. Exponentiation | Lambda NTTP / `std::function` | `IsInternalHom<A, B>` |
| 7. Equalizers | `dedekind::set<T, P>` | `IsEqualizer<f, g>` |
| 8. Empty Set | `dedekind::EmptySet` | `IsInitialObject<T>` |
| 9. Infinity (NNO) | `dedekind::Naturals` | `HasNaturalNumbersObject` |
| 10. Axiom of Choice | Lattice Dispatcher | `HasAxiom10ChoiceDispatcher` |

By adopting a categorical foundation, the library provides a unified interface for both *extensional* and *intensional* representations. For finite collections, such as the `Booleans`, the system

performs *extensional* materialization to compute exact results at compile-time. Conversely, for infinite or transcendental structures like the `Complex Numbers`, the library utilizes *intensional* symbolic expressions to maintain structural integrity. By embedding these categorical definitions in the `C++` host language, the programmer—and crucially the *compiler*—can examine the algebraic properties of a field even when the underlying elements cannot be fully materialized as discrete machine data.

### 4.2.6 Intensional Sets and Computational Undecidability

The utility of the categorical approach is most evident when defining sets whose membership conditions are conceptually clear but computationally undecidable. Consider the set of *Normal Numbers* $\mathscr{N}$, defined as those real numbers whose infinite digit expansion follows a uniform distribution. Formally, a real number $x$ is normal in base $b$ if for every finite sequence of digits $s$, the limiting frequency of $s$ appearing in the expansion of $x$ satisfies:

$$\lim_{n\to\infty} \frac{N(s,x,n)}{n} = \frac{1}{b^{|s|}} \tag{4.1}$$

where $N(s,x,n)$ denotes the number of occurrences of the string $s$ in the first $n$ digits of $x$. While we can easily define this property as a symbolic predicate within the `dedekind` ontology, the membership of specific constants like $\pi$ or $e$ remains a famous open problem in mathematics.

In a traditional imperative framework, such a set is unusable because membership cannot be *computed* to a boolean result. However, by treating $\mathscr{N}$ as an *intensional* set, the `dedekind` library allows the programmer to define morphisms that take $\mathscr{N}$ as their formal domain. This enables the `C++` compiler to verify the structural validity of a function pipeline based solely on the *definition* of normality, even while the empirical membership of $\pi$ remains unresolved. By hoisting the predicate into a `Stateless Constraint Type`, we decouple the logical requirement of the set from the necessity of materializing its elements. This allows the compiler to treat $\mathscr{N}$ as a valid algebraic object, facilitating symbolic reasoning about digit distribution without requiring a prior numerical proof of normality, thereby avoiding the runtime overhead of high-precision digit evaluation.

This allows the compiler to treat $\mathscr{N}$ as a valid algebraic object, facilitating symbolic reasoning about digit distribution without requiring a prior numerical proof of normality, thereby avoiding the runtime overhead of high-precision digit evaluation.

This intensional logic extends to non-elementary functions where the antiderivative cannot be expressed in a finite combination of elementary terms. For example, the Gaussian integral $e^{-t^2}$ defines the *Error Function* (erf) as a structural rule for area rather than a simple algebraic formula:

$$\operatorname{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} dt \tag{4.2}$$

While traditional backends rely on lossy polynomial approximations to evaluate this integral, the `dedekind` ontology reifies $\operatorname{erf}(x)$ as a set of symbolic properties. This allows the `C++` compiler to resolve identities—such as the symmetry $\operatorname{erf}(-x) = -\operatorname{erf}(x)$—as zero-overhead type transformations, delaying numerical approximation until the final realization of the continuum.

### 4.2.7 Categorification of Sequences and the Continuum

While ETCS provides the axioms for discrete collections, the representation of sequences and the analytic continuum requires a transition from set-theoretic membership to functorial structures. In the `dedekind` library, we avoid the imperative indexing of terms by treating sequences as *Combinatorial Species* [8]. By defining a sequence as a functor from the category of finite sets to itself, we can express growth rules—such as the Fibonacci recurrence—as algebraic identities between functors. This allows the compiler to resolve the structural identity of a series through

symbolic substitution before any machine-level summation occurs [9]. To reach the exact real arithmetic promised in Section 1, we ground our implementation of convergence in Lawvere's theory of enriched categories [38]. By treating metric spaces as categories enriched over the poset of non-negative reals $[0, \infty)$, the library reifies Cauchy sequences not as floating-point approximations, but as formal Cauchy completions [39]. This methodology ensures that the *Dedekind cut*—defining real numbers $\mathbb{R}$ as partitions of the rationals $\mathbb{Q}$ [34]—is not merely a runtime filter, but a verified limit of a rational sequence. This allows the `LLVM` backend to treat transcendental constants like $e$ and $\pi$ as symbolic invariants rather than opaque numeric variables.

The library addresses the representation of transcendental constants by distinguishing between their algebraic *definition* and their numerical *realization*. While the machine may be unable to *compute* the final digit of $\pi$, the `dedekind` ontology can lazily *define* it as a formal limit of a rational sequence. By encoding the Leibniz series for $\pi$ or the Taylor expansion for $e$, the library treats these constants as symbolic *intensional* types.

$$e = \sum_{n=0}^{\infty} \frac{1}{n!} = 1 + 1 + \frac{1}{2} + \frac{1}{6} + \dots \tag{4.3}$$

This approach leverages *lazy evaluation* at the type-level, where the `C++` compiler performs symbolic substitutions of the series terms before any floating-point approximation is required. By teaching the compiler the *rule* of the sequence rather than a static value, the library ensures that mathematical identities—such as $e^0 = 1$—can be resolved as zero-instruction compile-time proofs.

Table 4.2: Categorification of Sequences and Series in the Dedekind Ontology

| **Mathematical Concept** | **Set-Theoretic Form** | **Categorical Framework** | **C++ Structural** |
|---|---|---|---|
| **Recursive Sequence** | $F_n = F_{n-1} + F_{n-2}$ | Combinatorial Species [8] | Functorial Coprodu |
| **Formal Power Series** | $\sum a_n x^n$ | Analytic Functors [9] | Polynomial Species |
| **Metric/Distance** | $\lvert x - y \rvert < \epsilon$ | Lawvere Enriched Category [38] | Enriched Hom-sets |
| **Cauchy Convergence** | $\lim_{n\to\infty} a_n$ | Cauchy Completion [39] | Limit / Terminal O |
| **Real Numbers ($\mathbb{R}$)** | Dedekind Cuts / Cauchy | Archimedean Completion [21] | Adjoint Functors |

This mapping ensures that our software implementation is not merely a *simulation* of set theory, but a direct *instantiation* of its algebraic laws.

The structural mereology [22] approach in `dedekind` defines sets via formal relations—union ($X \cup Y$), intersection ($X \cap Y$), and parthood ($X \subseteq Y$) and last but not least element ($x \in Y$) – prior to implementation. This mirrors structuralist foundations such as Lawvere's Elementary Theory of the Category of Sets (ETCS) [40] and Cartesian Closed Categories (CCCs) [35, 41]. By prioritizing *structural transparency*, this methodology fundamentally departs from object-oriented encapsulation. In a sense, it encodes the well known rule of thumb of "composition over inheritance" [42]. In a standard `std::set`, the internal representation is an opaque implementation detail, invisible to the type system's logic. Conversely, in `dedekind`, the *type is the definition*. Because the set's constituents—predicates, universes, and mereological relations—are hoisted into the type signature as first-class members, the C++ type system and the `CMake` toolchain function as a Constraint Satisfaction Engine (CSE). This allows the compiler to inspect, decompose, and optimize a set based on its logical essence rather than its storage layout, trading the 'black-box' safety of encapsulation for the 'white-box' provability of formal mereology.

```
#include <concepts>
#include <type_traits>
```

```
// Verifies the "Element Of" relation (x in Y)
template <typename T, typename S>
concept IsElementOf = requires(T x, S s) {
    { s.\u03C7(x) } -> dedekind::category::LogicalValue;
};

// Verifies "Parthood" or "Subset" relation (X subset of Y)
template <typename Sub, typename Super>
concept IsSubsetOf = requires {
    requires std::derived_from<Sub, Super> ||
             requires { typename Sub::is_subset_of<Super>; };
};

// CSE Pruning: Collapsing contradictory intersections to EmptySet
template <typename SetA, typename SetB>
struct Intersection {
    using type = std::conditional_t<
        HasContradictoryPredicates_v<SetA, SetB>, // Evaluated at compile-time
        EmptySet,
        InternalIntersection<SetA, SetB>
    >;
};
```

Listing 4.1: Compile-time mereological verification using C++23 concepts.

By reifying a subset of the set-builder notation, dedekind turns the C++ translation unit into a symbolic laboratory. The programmer is free to reason in the language of Cantor, while the compiler produces the performance of a naked C++ loop.

## 4.3 Results

The `dedekind.sets` module is verified through the `sets_test` and `category_integration_test` suites. Table 4.1 maps the 10 ETCS axioms to their C++23 structural invariants. The module currently implements 8 of the 10 Lawvere axioms; Axioms 7 (unique factorisation through pullbacks) and 10 (well-pointedness) are deferred to future work as they require higher-order categorical machinery not yet integrated into the library.

# Chapter 5

# `dedekind.ieee` and `dedekind.numbers`

## 5.1 Introduction

The `dedekind.ieee` and `dedekind.numbers.approx` modules provide an algebraically-stamped IEEE 754 fast lane and a first-order error propagation framework. Standard floating-point arithmetic pervades scientific computing, yet its algebraic contract—associativity, commutativity, identity—is almost entirely implicit. These modules address this gap through an *effect-context* design: arithmetic remains fast by default, but every operation on the certified path is algebraically stamped and carries local sensitivity information.

Key references are the IEEE 754 standard [13], Higham's numerical stability guide [37], and error propagation literature [43, 44].

## 5.2 Implementation

### 5.2.1 The IEEE Fast Lane and Approximation Policies

**The `IEEE<F>` effect context.** The wrapper type `IEEE<F>` (where `F` is a `std::floating_point` type) acts as an opt-in semantic boundary. Entering the context via `assume_ieee(x)` asserts that the caller accepts IEEE 754 semantics [13]; exiting via `discharge_ieee(v)` recovers the raw value. The module-import boundary in C++23 plays a dual role: it is both a compilation unit edge and a conceptual quarantine zone that prevents unintentional widening of the certified region.

**Algebraic certification by policy fiat.** Rather than relying on the host language's implicit floating-point laws, the library explicitly registers algebraic structure for its operation tokens. `IEEEAdd<F>` and `IEEEMul<F>` are stateless tag types equipped with `SpeciesTraits` that declare Kleene-associativity and the appropriate identity element. These declarations are placed in the `dedekind::category` namespace so that the category-layer concept `IsAssociative<T, Op>`—defined in the `:total` partition (§ ?? if present)—can resolve for `T = IEEE<F>` without any further boilerplate at the call site.

**First-order error propagation (`dedekind.numbers.approx`).** The companion module supplies the `Approx<F>` envelope, which pairs a central value with a non-negative uncertainty radius. Given an operation token `Op`, the function

$$\texttt{propagate<F, Op>(a, b, op)} \tag{5.1}$$

computes the output value together with a first-order uncertainty bound derived from the Jacobian-style sensitivities provided by `Op::sensitivity(x, y)`. The propagation formula fol-

lows the classical treatment of Ku [43]:

$$\delta z \;\approx\; \left|\tfrac{\partial f}{\partial x}\right| \delta x + \left|\tfrac{\partial f}{\partial y}\right| \delta y, \tag{5.2}$$

with an additive machine-epsilon contribution modelled after the round-off proxy analysis of Higham [37]. This deliberately avoids the full interval-arithmetic cost [45] while still surfacing first-order conditioning information at run time.

**Propagation policies.** Three composable strategies govern how `Approx<F>` reacts when the computed uncertainty exceeds a configurable threshold:

- `IgnoreErrorPolicy`: discard the uncertainty tracker and return the raw IEEE value; maximises throughput for well-conditioned call sites.
- `ReportErrorPolicy`: preserve the `Approx<F>` result, exposing the central value together with its uncertainty radius so callers can inspect conditioning information without aborting the pipeline.
- `AdaptiveErrorPolicy`: inspect local cancellation diagnostics and, when the condition number indicates risk, prefer the safer association or factorisation [44].

All three policies are zero-overhead at compile time when the uncertainty radius is statically known to be zero.

**Concept-level contract: `IsIEEEPropagationOp<Op, F>`.** The constraint exported by `dedekind.numbers.ap` is grounded in the category layer rather than ad-hoc type traits:

```
export template <typename Op, typename F>
concept IsIEEEPropagationOp =
    std::floating_point<F> && IsAssociative<IEEE<F>, Op> &&
    requires(const Op& op, const IEEE<F>& x, const IEEE<F>& y) {
      { op({x, y}) } -> std::same_as<IEEE<F>>;
      { Op::sensitivity(x, y) } -> std::same_as<std::pair<F, F>>;
    };
```

Listing 5.1: Concept grounding the propagation contract on categorical associativity.

The `IsAssociative<IEEE<F>, Op>` clause delegates the associativity proof to the same concept that governs all algebraic structures in the library, ensuring that the numeric fast lane is not an isolated corner of the type hierarchy but a fully categorified citizen of the `dedekind` ontology.

## 5.3 Results

### 5.3.1 Fast Lane vs. Improved Numerics

Table 5.1 summarizes the empirical design contrast between the IEEE "fast lane" (direct machine evaluation) and the improved numerics path (adaptive/reporting policies with first-order propagation). The key point is not to replace the fast lane, but to make it inspectable: high-throughput execution remains the default, while difficult terrain can be detected and selectively rerouted using local sensitivity/cancellation signals.

In our current implementation, adaptive checks are deliberately local (operation-neighborhood scope), which keeps dispatch predictable and avoids global symbolic explosion. This follows standard numerical analysis guidance: combine inexpensive local conditioning signals with targeted rewrites, and only escalate to expensive methods when diagnostics indicate instability [37].

**Showcase: Computing with Infinities**

Table 5.1: Comparison of IEEE fast lane and improved numerics policies in `dedekind`.

| **Aspect** | **Fast lane (IEEE default)** | **Improved numerics (adaptive/reporting)** |
|---|---|---|
| Primary objective | Maximal throughput with direct IEEE operations | Preserve throughput while attaching local error/safety context |
| Semantic contract | Accepts non-associativity and special values as machine semantics | Classifies outcomes (Regular/NearZero/Huge/NonFinite) and reports confidence |
| Error model | Implicit, usually post-hoc debugging | Explicit first-order propagation via Jacobian-style sensitivities |
| Typical overhead | Minimal | Low-to-moderate metadata overhead; no mandatory arbitrary precision |
| Corner-case handling | Relies on caller discipline | Policy-level response (Ignore/Report/Adaptive) and rewrite hints |
| Optimization posture | Compiler may reassociate only where laws are certified | Adaptive path can prefer safer association/factorization when cancellation risk is high |
| Theoretical anchors | IEEE floating-point standard [13] | Stability and propagation literature [37, 43, 45, 44] |

```
#include <dedekind/sets.hpp>
#include <dedekind/numbers.hpp>

using namespace dedekind::category;

// 1. Define an intensional set: All Prime Numbers > 2
// In dedekind, this is a Stateless Constraint Type
using PrimesGt2 = Set<int, [](int n) {
    return is_prime(n) && (n > 2);
}>;

// 2. Define another intensional set: All Even Numbers
using Evens = Set<int, [](int n) {
    return (n % 2 == 0);
}>;

// 3. Symbolic Intersection
// The compiler inspects the 'SpeciesTraits' and predicates
using ForbiddenZone = Intersection<PrimesGt2, Evens>;

// --- THE "IMPRESSIVE" BIT ---

// A C++ developer expects this to require a runtime check or loop.
// Instead, dedekind performs "Structural Pruning".
```

Listing 5.2: Showing that the intersection of two non-overlapping sequences is the empty set.

**Showcase: Simple Differential Equations**

```
#include <dedekind/analysis.hpp>
```

```
// Define 'e^x' as an intensional symbolic type
using Exp = Symbolic::Exponential;

// The 'Derivative' meta-function computes the symbolic result
using ExpPrime = Derivative<Exp>;

// The compiler 'solves' the equation by confirming identity.
static_assert(std::is_same_v<Exp, ExpPrime>,
    "Proof: f' = f is satisfied by the Exponential species.");

// Usage in a 'Highway' pipeline
void calculate() {
    // Because the solution is proven at compile-time,
    // the machine code is just a literal constant loading.
    constexpr auto result = Exp::at(0); // result = 1.0
}
```

Listing 5.3: Compile-time solution of a simple differential equation.

| **Structure** | **Categorical Type** | **Unit ($\eta$)** | **Mult. ($\mu$)** | **Symmetry** |
|---|---|---|---|---|
| `std::identity_t` | Identity Monad | `x` | `x` | Trivial |
| `std::optional` | Maybe Monad ($A+1$) | `just(x)` | `flatten` | N/A |
| `std::variant` | Coproduct Monad ($\sum A_i$) | `in_i(x)` | `join` | Frobenius |
| `std::vector` | List Monad ($A^*$) | `{x}` | `concat` | Frobenius |
| `std::ranges::view` | Stream Monad | `single(x)` | `join` | Frobenius |
| `std::pair<W, A>` | Writer / Product | `{i_w, x}` | `combine_w` | Adjoint |

Table 5.2: C++23 Standard Types as Endofunctors with Frobenius Symmetry

- **Ranges as Frobenius Algebras:** `std::ranges::view` is identified as a Frobenius object in the category of species. It satisfies the monadic requirements for non-deterministic computation (*morphism-lift* via `transform` and *multiplication* via `join`) and the comonadic requirements for local context extension (*comultiplication* via `tails/windows` and *counit* via `front`).
- **Highway Duality:** Within the `dedekind` ontology, the `operator»` (Forward Highway) executes the monadic bind, while `operator«` (Upstream Highway) executes the comonadic extension, providing a symmetric interface for range-based pipelines.

Table 5.3: The Dedekind Registry: Mapping C++23 Standard Primitives to Categorical Species.

| **Mathematical Concept** | **Standard C++23 Entity** | **Dedekind Concept** | **Ontological Status** |
|---|---|---|---|
| *Boundary Objects (Limits)* | | | |
| Terminal Object (1) | `std::monostate` | `IsTerminalObject` | **Total** (Singleton) |
| Initial Object (0) | `std::nullptr_t` | `IsInitialObject` | **Total** (Unreachable) |
| *Morphic Foundations* | | | |
| Strict Product ($A \times B$) | `std::pair<A, B>` | `IsProduct` | **Strict** (Non-conv.) |
| Coproduct ($A + B$) | `std::variant<A, B>` | `IsCoproduct` | Rule of Choice |
| *Function Spaces (Exponentials)* | | | |
| Internal Hom-set ($B^A$) | `std::move_only_function<B(A)>` | `IsExponential` | **Move-Only** (Linear) |
| Evaluation Morphism ($\epsilon$) | `std::invoke` | `eval` | **Universal** |
| Abstraction (Currying) | `[=](A a) { return f(x, a); }` | `IsExponential` | **Adjunction** |
| Structural Closure | `[cap](A a) { ... }` | `IsExponential` | **Structural** (Anonymous) |
| *Algebraic Species (Total)* | | | |
| Modular Group ($\mathbb{Z}/2^n\mathbb{Z}, +$) | `unsigned int`, `std::plus` | `IsAbelianGroup` | **Total** (Closed) |
| Boolean Rig ($\mathbb{B}, \vee, \wedge$) | `bool`, `\|\|`, `&&` | `IsRig` | **Total** (Idempotent) |
| Modular Ring ($\mathbb{Z}/2^n\mathbb{Z}, +, \times$) | `unsigned int`, `+`, `*` | `IsRing` | **Total** (Distributive) |
| *Hazardous Species (Rejected)* | | | |
| Hazardous Magma | `int`, `std::plus` | `IsPointed` | **Hazard** (UB Overflow) |
| Floating Point Magma | `double`, `std::plus` | `IsMagma` | **Hazard** (Non-assoc.) |
| *Relational Lattices* | | | |
| Distributive Lattice | `bool`, `std::logical_or` | `IsDistributiveLattice` | **Total** |
| Pre-order ($\sqsubseteq$) | `operator<=` | `IsPartOf` | Mereological Basis |

To conclude the implementation hierarchy, we provide a formal mapping between the abstract nomenclature of the mathematical literature and the concrete syntax of the `C++23` language. In this structuralist posture, the "identity" of a species is not found in its nominal inheritance, but in its ability to satisfy the mereological and algebraic requirements dictated by the **Dedekind Atlas** (see Table 5.4).

By anchoring established axioms directly into the standard operator set—where `operator<=` serves as the formal proof of Parthood ($\sqsubseteq$) and the "Highway" `operator»` represents Kleisli composition—we ensure that the resulting system is not merely a simulation of a mathematical model, but a direct execution of its laws. This registry serves as the final "Seal of Truth" for the implementation, providing the compiler with the semantic map required to navigate the ontology.

Table 5.4: The Dedekind Semantic Mapping: A Registry of Formal Correspondences

| **Mathematical Concept** | **Symbol** | **C++ Operator / Type** | **C++ Concept** | **Basis** |
|---|---|---|---|---|
| *Category Theory* | | | | |
| Kleisli Extension | $(T, \eta, \gg=)$ | `operator»=` | `IsKleisliExtension` | `IsAssociative` |
| Kleisli Extension | $\gg=$ | `operator»=` | `IsKleisliExtension` | `IsAssociative` |
| Kleisli Composition | $\circ\_K$ | `operator»` | `IsMonad` | `IsAssociative` |
| Co-Kleisli Extension | $\gg=_{co}$ | `operator«=` | `IsComonad` | `IsAssociative` |
| Morphic Injection (Unit) | $\eta$ | `into<F>` / `singleton` | `IsMonad` | $\eta : T \to F(T)$ |
| Morphic Extraction (Counit) | $\varepsilon$ | `extract<F>` | `IsComonad` | $\varepsilon : F(T) \to T$ |
| Characteristic Morphism | $\chi$ | `operator()` | `IsCharacteristic` | |
| Co-Kleisli Composition | $\circ\_CoK$ | `operator«` | `IsComonad` | |
| Counit (Extraction) | $\varepsilon$ | `extract<F>` | `IsComonad` | |
| *Mereology* | | | | |
| Proper Parthood | $\chi$ | `operator()` | `IsProperPart` | `IsCharacteristic` |
| Parthood (Pre-order) | $\sqsubseteq$ | `operator<=` | `IsPartOf` | |
| Mereological Sum (Join) | $\sqcup$ | `operator\|` | `IsJoinSemilattice` | |
| Mereological Product (Meet) | $\sqcap$ | `operator&` | `IsMeetSemilattice` | |
| Universal Complement | $\neg$ | `operator!` | `IsComplemented` | |
| Initial Object (Empty) | $\emptyset$ | `∅<T>` | `IsInitialObject` | |
| Terminal Object (Universal) | $\Omega$ | `Ω<T>` | `IsTerminalObject` | |
| *Set Theory* | | | | |
| Membership | $\in$ | `operator()` | `IsSet` | `IsProperPart` |
| Subset | $\subseteq$ | `operator<=` | `IsSubset` | `IsPartOf` |
| Countable Infinity | $\aleph_0$ | `ℵ_0` | `IsCardinality` | |
| Continuum Cardinality | $\beth_1$ | `ℶ_1` | `IsCardinality` | |
| *Order Theory* | | | | |
| Pre-order | $\sqsubseteq$ | `operator<=` | `IsPartiallyOrdered` | `IsPartOf` |
| *Algebra* | | | | |
| Structural Origin (Zero) | $0$ | `T::origin()` | `IsPointed` | |
| Additive Composition | $+$ | `operator+` | `IsGroup` | |
| Additive Inverse | $-$ | `operator-` | `IsGroup` | |
| Multiplicative Composition | $\times$ | `operator*` | `IsRing` | |
| Multiplicative Inverse | $/, {}^{-1}$ | `operator/` | `IsField` | |
| Associativity | $(a \circ b) \circ c$ | `is_associative_v` | `IsSemigroupoid` | |
| Idempotency | $a \circ a = a$ | `is_idempotent_v` | `IsIdempotent` | |
| Magnitude / Count | $\|S\|$ | `s.size()` / `s.cardinality()` | `IsCardinality` | |
| Supremum / Infimum | $\sup, \inf$ | `s.supremum(), s.infimum()` | `HasExtrema` | |

# Chapter 6

# dedekind.order

## 6.1 Introduction

The `dedekind.order` module constitutes *Level 1.5* of the ontological hierarchy, establishing the rules of relative positioning before introducing cardinality or measure. Order is the prerequisite for limits and convergence: one cannot define a Cauchy sequence without first knowing what it means for two elements to be comparable.

Key references: Davey & Priestley [46], Birkhoff [47], and the general treatment in Pierce [25].

## 6.2 Implementation

The module is a single-partition layer exposing four graduated order concepts, each a conservative extension of the last.

Table 6.1: Rosetta Stone: `dedekind.order`

| **Mathematical Concept** | **C++23 construct** | **Structural `concept`** |
|---|---|---|
| Pre-order ($\leq$, reflexive + transitive) | `operator<=` | `IsPreOrdered<T>` |
| Directed set (join-semilattice) | `operator\|` | `IsDirectedSet<T>` |
| Partial order (+ antisymmetry) | `std::partial_ordering` | `IsPartiallyOrdered<T>` |
| Total order (chain, no parallelism) | `std::totally_ordered<T>` | `IsTotallyOrdered<T>` |

```
import dedekind.order;
using namespace dedekind::order;

// The integer chain satisfies all four levels
STATIC_CHECK(IsPreOrdered<int>);
STATIC_CHECK(IsPartiallyOrdered<int>);
STATIC_CHECK(IsTotallyOrdered<int>);

// Booleans form a poset (total order requires a witness)
STATIC_CHECK(IsPartiallyOrdered<bool>);
```

Listing 6.1: Order: compile-time concept verification (from `order_test.cpp`).

## 6.3 Results

The module validates that every concrete arithmetic species registered in `dedekind.category` satisfies at least `IsPreOrdered`. Machine integers pass all four levels, confirming their certification as fully ordered chains.

# Chapter 7

# dedekind.topology

## 7.1 Introduction

The `dedekind.topology` module constitutes *Level 2* of the hierarchy, encoding the notion of "closeness" without requiring a full metric. It refines the sets of `dedekind.sets` with open-set structure through two geometric primitives: half-spaces (rays) and bounded neighbourhoods (intervals).

Key references: Munkres [48], and the categorical treatment in Lawvere & Schanuel [34].

## 7.2 :interval

An *Interval* is the canonical intersection of two opposing half-spaces (rays). It is the prototypical open neighbourhood of a point in an ordered space.

Table 7.1: Rosetta Stone: `dedekind.topology`

| **Mathematical Concept** | **C++23 construct** | **Structural** `concept` |
|---|---|---|
| Half-space / ray $(x > a)$ | `Ray<T, Direction>` | `IsHalfSpace<T>` |
| Open interval $(a, b)$ | `Interval<T>` | `IsInterval<T>` |
| Convex set | `operator&` on rays | `IsConvex<T>` |
| Open set | `is_open_tag` | `IsOpen<T>` |
| Neighbourhood of $p$ | open set containing $p$ | `IsNeighborhood<T,P>` |

```
using ℝ = int;
using UnitInterval = Interval<ℝ>;
using UnitRay      = Ray<ℝ, Direction::Upward>;

static_assert(IsOpen<UnitRay>);
static_assert(IsOpen<UnitInterval>);
static_assert(IsConvex<UnitRay>);
static_assert(IsConvex<UnitInterval>);
```

Listing 7.1: Topology: open sets and convexity (from `shapes_test.cpp`).

## 7.3 :neighborhood

A *Neighbourhood* of a point $p$ is an open set that contains $p$. This partition encodes the local structure needed to define continuity and limits without an explicit metric.

```
UnitInterval neighborhood(0, 2);
ℝ point = 1;

static_assert(IsNeighborhood<UnitInterval, ℝ>);
REQUIRE(neighborhood(point) == ClassicalLogic::True);
```

Listing 7.2: Topology: neighbourhood membership (from `shapes_test.cpp`).

## 7.4 Results

Both `Ray` and `Interval` satisfy the full topology concept stack, confirming that the geometric layer provides a verified foundation for limits in `dedekind.sequences`.

# Chapter 8

# dedekind.sequences

## 8.1 Introduction

The `dedekind.sequences` module constitutes *Level 2.5* of the hierarchy, providing the formal machinery for convergence. Built upon the topology of `dedekind.topology`, it extends the category of paths to Cauchy sequences and nets, enabling the formal treatment of limits without presupposing a complete metric space.

Key references: the categorical treatment via Borceux & Dejean [39] and Lawvere's metric-space characterisation [38].

Table 8.1: Rosetta Stone: `dedekind.sequences`

| **Mathematical Concept** | **C++23 construct** | **Structural** `concept` |
|---|---|---|
| Net / sequence | `Path<T>` | `IsSequence` |
| Cauchy criterion | point-wise $\lvert s_m - s_n \rvert$ | runtime test |
| Comonadic extension | `operator<<=` | comonad `extend` |
| Convergent limit | `:limits` partition | `IsConvergent` |

## 8.2 :net

A *Net* is the most general notion of convergence, indexed by a directed set. This partition provides the `IsSequence` concept as the foundational structural requirement.

## 8.3 :limits

The `:limits` partition provides Cauchy-convergence infrastructure: the criterion that for any $\varepsilon > 0$ there exists $N$ such that $m, n > N \Rightarrow |s_m - s_n| < \varepsilon$.

```
Path<double> harmonic{
    [](std::size_t n) { return 1.0 / static_cast<double>(n + 1); }};

static_assert(IsSequence<decltype(harmonic)>);

// Cauchy criterion: distant terms are close
double s_1000 = harmonic.at(1000);
double s_2000 = harmonic.at(2000);
REQUIRE(std::abs(s_1000 - s_2000) < 0.001);
```

Listing 8.1: Sequences: Cauchy convergence of the harmonic series (from `cauchy_test.cpp`).

## 8.4 :path

`Path<T>` is a comonadic container: it supports sampling via `.at(n)`, and comonadic extension (`operator<<=`) for contextual operations over neighbouring terms.

```
using ℤ = int;
Path<ℤ> path{[](std::size_t n) { return static_cast<ℤ>(n) + 42; }};

REQUIRE(path.at(0) == 42);

// Comonadic extension: first differences
auto diffs = path <<= [](const Path<ℤ>& ctx) {
    return ctx.at(0) - ctx.at(1);
};
REQUIRE(diffs.at(0) == -1);
```

Listing 8.2: Sequences: comonadic path extension (from `path_test.cpp`).

## 8.5 Results

`Path<T>` passes `IsSequence` and the Cauchy criterion for the harmonic series, confirming the sequences layer as a verified foundation for the continuum in `dedekind.numbers`.

# Chapter 9

# dedekind.algebra

## 9.1 Introduction

The `dedekind.algebra` module builds upon `dedekind.sets` to implement rings, fields, semimodules, and modules via reduced-strength algebraic structures. A core design principle is to avoid the convenient `double` route and instead spell out the algebraic structure explicitly: from boolean rigs through unsigned-integer rings up to the complex-number field. A recurring implementation challenge was the pressure to default to `double`; the module systematically resisted this in favour of explicit semimodules and semirings.

The name "algebra" derives from al-Khwārizmī's *al-Kitāb al-mukhtasar fī hisāb al-jabr wal-muqābala* (820 CE).

Table 9.1: Rosetta Stone: `dedekind.algebra`

| **Mathematical Concept** | **C++23 construct** | **Structural `concept`** |
|---|---|---|
| Monoid $(S, *, e)$ | `std::multiplies` / `std::plus` | `IsAdditiveMonoid`, `IsMulMonoid` |
| Group $(S, *, e, \text{inv})$ | unary `operator-` | `IsAdditiveGroup` |
| Ring $(S, +, \cdot)$ | `operator+` × `operator*` | `IsRing` |
| Field (ring + inverses) | `.inverse()` | `IsField` |
| Module ($R$-action on $S$) | scalar `operator*` | `IsModule` |
| Polynomial ring $R[x]$ | `Polynomial<R>` | `IsPolynomial` |
| Boolean rig | `bool` / `IsBooleanAlgebra` | `IsBooleanRig` |

## 9.2 :boolean

Encodes Boolean algebras as a reduced-strength rig (semi-ring without negation) that lifts the truth-values of `ClassicalLogic` into algebraic form.

## 9.3 :group

Certifies additive and multiplicative groups. Machine integers are deliberately *not* certified as additive groups due to overflow semantics.

```
int q = 42, zero = 0;
CHECK(q + zero == q);       // identity
CHECK(q + (-q) == zero);    // inversion
```

Listing 9.1: Algebra: group identity and inversion (from `group_test.cpp`).

## 9.4 :ring

The ring is the fundamental compound structure: an additive group with a distributive multiplication.

```
int a = 6, b = 7;
CHECK(a * b == 42);
CHECK(std::multiplies<int>{}(a, b) == 42);
```

Listing 9.2: Algebra: ring multiplication (from `ring_test.cpp`).

## 9.5 :field

A field is a ring where every nonzero element has a multiplicative inverse. The canonical implementation is `Rational<Z>` from `dedekind.numbers`.

## 9.6 :division

Provides the division morphism $a/b = a \cdot b^{-1}$ and the `IsDivisionRing` concept as a precursor to fields.

## 9.7 :modules

Semimodules and modules formalise the action of a ring on an additive abelian group, generalising vector spaces to arbitrary coefficient rings.

## 9.8 :polynomial

`Polynomial<R>` encodes the free algebraic object $\sum_k a_k x^k$ over a ring $R$, providing the carrier for symbolic manipulation.

## 9.9 Results

The algebra module validates that `int` satisfies ring axioms and that `Rational<int>` satisfies field axioms, bridging the abstract categorical layer and the concrete numeric tower.

# Chapter 10

# dedekind.morphologies

## 10.1 Introduction

The `dedekind.morphologies` module constitutes *Level 3.5* of the hierarchy, encoding the structural patterns that constrain how algebraic species behave globally. A morphology is a global property—the absence of infinitely large elements (Archimedean), or the periodicity of the successor map (cyclic groups)—rather than an operation.

Table 10.1: Rosetta Stone: `dedekind.morphologies`

| **Mathematical Concept** | **C++23 construct** | **Structural** `concept` |
|---|---|---|
| Archimedean property | trait witness | `IsArchimedean<T>` |
| Cyclic group $\mathbb{Z}/n\mathbb{Z}$ | `CyclicRing<T, N>` | `IsCyclic`, `IsCyclicRing` |
| Successor chain | `CyclicRing<T,N>::successor` | wrap-around at $N$ |

## 10.2 :archimedean

Certifies that a species has no infinitesimal or infinite elements relative to its unit: for all $x, y > 0$ there exists $n \in \mathbb{N}$ such that $n \cdot x > y$. This is a prerequisite for the construction of Dedekind cuts in `dedekind.numbers`.

## 10.3 :cyclic

`CyclicRing<T, N>` implements the Dedekind–Peano successor structure for finite rings, with a compile-time verified wrap-around at $N$.

```
using ℤ100 = CyclicRing<int, 100>;

static_assert(IsCyclic<ℤ100>);
static_assert(IsCyclicRing<ℤ100>);

// Successor wraps at boundary
static_assert(ℤ100::successor(ℤ100{99}) == 0);

// Modular arithmetic: 70 + 50 = 20  (mod 100)
constexpr ℤ100 a{70}, b{50};
static_assert(static_cast<int>(a + b) == 20);
```

Listing 10.1: Morphologies: cyclic ring wrap-around (from `cyclic_test.cpp`).

## 10.4 Results

Compile-time checks confirm that `CyclicRing<int, 100>` satisfies both `IsCyclic` and `IsCyclicRing`, and that modular arithmetic wraps correctly at the ring boundary.

# Chapter 11

# dedekind.geometry

## 11.1 Introduction

The `dedekind.geometry` module extends the algebraic module structure with geometric operations: displacement, inner products, norms, and completeness. It provides `Vector<S, N>` as the canonical $N$-dimensional vector species over a scalar ring $S$.

Table 11.1: Rosetta Stone: `dedekind.geometry`

| **Mathematical Concept** | **C++23 construct** | **Structural** `concept` |
|---|---|---|
| Affine / vector space | `Vector<S,N>` | `IsAffine` |
| Inner product $\langle u, v \rangle$ | `dot(u, v)` | `IsInnerProductSpace` |
| Normed space $\|v\|$ | `norm(v)` | `IsNormedSpace` |
| Hilbert space (complete) | real $\mathbb{R}^N$ | `IsHilbertSpace` |

## 11.2 :affine

Implements $N$-dimensional vectors with component-wise arithmetic. The affine layer provides translation and scalar action without presupposing a norm.

```
using Vec2 = Vector<double, 2>;
Vec2 v{1.0, 1.0};
auto res = v * 0.5;          // scalar action

Vec2 a{0.5, 0.0}, b{0.5, 1.0};
auto c = a + b;              // vector addition
REQUIRE(c[0] == 1.0);
```

Listing 11.1: Geometry: affine vector operations (from `affine_test.cpp`).

## 11.3 :inner_product

Defines the bilinear pairing $\langle -, - \rangle$ and the derived notion of orthogonality ($\langle u, v \rangle = 0$).

## 11.4 :euclidean

Adds the $L^2$ norm $\|v\| = \sqrt{\langle v, v \rangle}$ and the triangle inequality, establishing the metric structure required for geometric limits.

## 11.5 :hilbert

Certifies completeness: every finite-dimensional real vector space is trivially complete. `IsHilbertSpace` encodes this as a compile-time guarantee.

```
using Vec3 = Vector<double, 3>;

Vec3 x{1.0, 0.0, 0.0}, y{0.0, 1.0, 0.0};
REQUIRE(dot(x, y) == 0.0);          // orthogonality

Vec3 v{3.0, 4.0, 0.0};
REQUIRE(norm(v) == 5.0);             // L2 norm == 5

static_assert(IsHilbertSpace<Vec3, double>);
```

Listing 11.2: Geometry: Hilbert space verification (from `hilbert_test.cpp`).

## 11.6 Results

`Vector<double, N>` satisfies the full four-level geometry stack, confirming it as both an inner-product and a Hilbert space.

# Chapter 12

# dedekind.numbers

## 12.1 Introduction

The `dedekind.numbers` module is the top-level integration layer for the numeric tower. It provides embedding morphisms between all numeric species: $\mathbb{B} \hookrightarrow \mathbb{N} \hookrightarrow \mathbb{Z} \hookrightarrow \mathbb{Q} \hookrightarrow \mathbb{R} \hookrightarrow \mathbb{C} \hookrightarrow \mathbb{D}$ (dual numbers), with each embedding preserving the algebraic structure certified by the downstream modules.

Table 12.1: Rosetta Stone: `dedekind.numbers`

| **Mathematical species** | **C++23 type** | **Embedding** |
|---|---|---|
| $\mathbb{B}$ (Booleans) | `bool` / `Truth<L>` | $\hookrightarrow \mathbb{N}$ via `embed_B_N` |
| $\mathbb{N}$ (naturals) | `unsigned` | $\hookrightarrow \mathbb{Z}$ via `embed_N_Z` |
| $\mathbb{Z}$ (integers) | `int` / `machine_integer` | $\hookrightarrow \mathbb{Q}$ via `embed_Z_Q` |
| $\mathbb{Q}$ (rationals) | `Rational<Z>` | $\hookrightarrow \mathbb{R}$ (cut) |
| $\mathbb{R}$ (reals) | Dedekind cut / `approx` | $\hookrightarrow \mathbb{C}$ |
| $\mathbb{C}$ (complex) | `Complex<S>` | Cayley–Dickson |
| $\mathbb{D}$ (dual) | `Dual<F>` | automatic differentiation |

## 12.2 :booleans

`Truth<L>` lifts the truth-values of a logic $L$ into the numerical species hierarchy, providing the base of the embedding chain.

## 12.3 :naturals

Wraps `unsigned` as the canonical $\mathbb{N}$. The embedding *embed*_$\mathbb{B}$_$\mathbb{N}$ sends `false` $\mapsto 0$, `true` $\mapsto 1$.

## 12.4 :integer

Wraps signed machine integers as $\mathbb{Z}$ with the canonical embedding *embed*_$\mathbb{N}$_$\mathbb{Z}$.

## 12.5 :rational

`Rational<Z>` implements $\mathbb{Q}$ as a GCD-normalised fraction, satisfying the full field axioms.

```
Rational<int> q(12, 18);    // GCD(12,18)=6
REQUIRE(q.num() == 2);
REQUIRE(q.den() == 3);

Rational<int> a(3, 4);
auto identity = a * a.inverse();  // = 1/1
REQUIRE(identity.num() == 1);
```

Listing 12.1: Numbers: rational simplification and field inverse (from `rational_test.cpp`).

## 12.6 :complex

`Complex<S>` implements $\mathbb{C}$ via the Cayley–Dickson construction from any scalar ring.

## 12.7 :scalars

Provides the floating-point anchors `machine_real_scalar` and `machine_integer` as pre-certified concrete species.

## 12.8 :dual

`Dual<F>` implements dual numbers $a + b\varepsilon$ ($\varepsilon^2 = 0$) for automatic differentiation. The value component `.value()` is the primal; `.derivative()` is the tangent.

## 12.9 :symbolic

Formal symbols (indeterminates) for polynomial and symbolic computation. Provides the carrier type for `Polynomial<R>` in `dedekind.algebra`.

## 12.10 :real

The real-number construction via Dedekind cuts: a real number is a partition $(L, U)$ of $\mathbb{Q}$ such that $L$ is non-empty, downward-closed, and has no maximum.

## 12.11 :constants

Named transcendental constants $\pi$, $e$, $\gamma$, etc., certified as Dedekind cuts over $\mathbb{Q}$.

```
// Set membership via embedding chain
const auto naturals = ambient_set<unsigned>(
    [](const unsigned&) { return true; });
CHECK(in_via(true,  embed_B_N, naturals) == true);
CHECK(in_via(false, embed_B_N, naturals) == true);

// Compose the full chain into Q+ predicate
const auto embed_B_Q = [](bool b) {
    return embed_Z_Q<>(embed_N_Z(embed_B_N(b)));
};
const auto pos = [](const Rational<machine_integer>& q) {
    return q.num() > 0;
};
const Set<Rational<machine_integer>,
          ClassicalLogic, decltype(pos)> Q_plus{pos};
```

```
CHECK(Q_plus(embed_B_Q(true))  == true);    // 1/1 ∈ Q+
CHECK(Q_plus(embed_B_Q(false)) == false);   // 0/1 ∉ Q+
```

Listing 12.2: Numbers: inter-species embedding $\mathbb{B} \hookrightarrow \mathbb{N} \hookrightarrow \mathbb{Z} \hookrightarrow \mathbb{Q}$ (from `tower_test.cpp`).

## 12.12 Results

The physical build order of the `dedekind` module graph mirrors the conceptual hierarchy. By enforcing a linear dependency graph, the compiler's proof-search for a Ring's identity element is grounded in the prior verification of its mereological subobjects. All embedding morphisms are verified by the compiler as monic arrows, and the tower test suite validates end-to-end membership across species boundaries.

# Chapter 13

# dedekind.analysis

## 13.1 Introduction

The `dedekind.analysis` module occupies the apex of the build hierarchy, synthesising geometry, morphisms, and differential structure into the formal machinery of classical and functional analysis. It provides differential $k$-forms, the Hamilton–Jacobi framework, and kernel (neural network) primitives.

Table 13.1: Rosetta Stone: `dedekind.analysis`

| **Mathematical Concept** | **C++23 construct** | **Structural** `concept` |
|---|---|---|
| Differential $k$-form $\omega$ | `OneForm<S,N>` / `TwoForm` | `IsForm` |
| Wedge product $\alpha \wedge \beta$ | `wedge(\alpha, \beta)` | antisymmetric bilinear map |
| Hamiltonian flow | `::hamilton` partition | area-preserving action |
| Kernel / activation | `::kernels` partition | `IsKernel` |

## 13.2 :exterior

Implements the exterior algebra of differential forms. The wedge product $\alpha \wedge \beta$ is the antisymmetric tensor product, essential for Hamiltonian mechanics and integration theory.

## 13.3 :forms

Provides `OneForm<S,N>` (co-vectors / linear functionals) and the induced `TwoForm` (area element), the building blocks for symplectic geometry.

```
using Vec2 = Vector<double, 2>;
OneForm<double, 2> dq{{1.0, 0.0}}, dp{{0.0, 1.0}};
auto omega = wedge(dp, dq);    // symplectic form

Vec2 u{1.0, 0.0}, v{0.0, 1.0};
REQUIRE(omega(u, v) == -1.0);  // orientation
REQUIRE(omega(v, u) ==  1.0);  // antisymmetry
```

Listing 13.1: Analysis: symplectic 2-form and area preservation (from `hamilton_test.cpp`).

## 13.4 :hamilton

The Hamiltonian partition encodes phase-space flows preserving the symplectic form $\omega = dp \wedge dq$, i.e. Liouville's theorem in the language of exterior algebra.

```
// Simple harmonic oscillator: H = 0.5*(p^2 + q^2)
auto H = [](auto state) {
    auto q = state[0], p = state[1];
    return 0.5 * (p*p + q*q);
};
Vec2 state_t0{1.0, 0.0}, state_t1{0.0, 1.0};
// Area in phase space is preserved
REQUIRE(std::abs(omega(state_t0, state_t1)) == 1.0);
```

Listing 13.2: Analysis: Hamiltonian area preservation (from `hamilton_test.cpp`).

## 13.5 :kernels

Kernel functions $k(x, y) = \langle \phi(x), \phi(y) \rangle$ encode feature maps into Hilbert spaces, providing the theoretical substrate for Support Vector Machines and neural network activation layers.

## 13.6 Results

The analysis module validates the antisymmetry of the wedge product and the area-preserving property of Hamiltonian flows, confirming that the library's apex layer correctly materialises classical mechanics in C++23 type theory.

# Chapter 14

# Benchmarks and Structural Collapse

## 14.1 Performance

**Computational Complexity of the Build Graph:** While template metaprogramming is Turing-complete and prone to pathological instantiation depth, the `dedekind` library utilizes C++23 Module Partitions to enforce a strict Directed Acyclic Graph (DAG) of truth. In practice, we observe that build times follow an $O(L)$ complexity, where $L$ is the depth of the partition in the ontological hierarchy.

- **Incremental Stability:** Changes to terminal partitions (e.g., `:numeric`) result in $O(1)$ rebuild times, as the foundational Binary Module Interfaces (BMIs) for `:morphism` and `:small` remain cached.
- **Linear Scaling:** A full cold-start rebuild scales as $O(N)$, where $N$ is the number of partitions, preventing the exponential "template blow-up" typically associated with header-only functional C++ libraries.

This stratification ensures that the formal verifiation of the continuum does not destabilize the build pipeline, making Axiomatic Systems Programming viable for industrial-scale symbolic computing. Exponential build times, while strictly speaking possible, have not been observed during library development.

**3.9 Benchmarks: The Pragmatic Displacement.** We evaluate `dedekind` against standard implementations from the Computer Language Benchmarks Game. Our goal is to measure the "template tax" of Axiomatic Systems Programming relative to both procedural C++ and high-level Python backends.

- **pidigits:** By modeling $\pi$ as a formal limit of a rational sequence, we compare the resolution of transcendental numbers against the CPython `gmpy2` wrapper. We demonstrate that symbolic lazy evaluation in `dedekind` maintains a zero-overhead profile relative to raw C/GMP.
- **spectral-norm:** We reify the infinite matrix $A$ as a combinatorial species. This benchmark serves as a stress test for the LLVM backend's ability to perform structural pruning across Highway-composed operator chains, targeting bit-perfect parity with hand-tuned assembly.

## 14.2 The Sieve of Dedekind: Existential Proof of Structural Collapse

To bridge the semantic gap between high-level mathematical intent and machine execution, we provide an existential proof of *Structural Collapse*. While general predicates can lead to com-

putational undecidability at compile-time, `dedekind` implements a pragmatic "Winning Move" by geometrizing the type system through the use of **Half-Spaces** and **Intervals**.

In this architecture, sets are reified as canonical geometric primitives. For "well-behaved" species—specifically the rings of integers ($\mathbb{Z}$) and rationals ($\mathbb{Q}$) where equality and order are decidable—the intersection of two sets is transformed into a **Static Feasibility Check**.

```
// Geometrizing the Type System: Half-Spaces in Z

// The Symbolic Sieve: The compiler resolves the intersection

// Existential Proof: The following block produces ZERO machine instructions
}
```

Listing 14.1: Symbolic reduction of contradictory half-spaces into a no-op.

### 14.2.1 Pragmatic Auditing via the CI Pipeline

As a prerequisite for publication, we leverage the `dedekind` CI pipeline as an **Independent Auditor** to verify this "all-or-nothing" optimization. Unlike traditional runtime benchmarks, our methodology focuses on **IR-Level Verification**:

- **Sentinel Tracking:** We embed assembly markers (sentinels) within the lowering path of symbolic operations. The CI pipeline audits the LLVM Intermediate Representation (IR) to confirm that contradictory intersections are physically elided.
- **Static Sieve Verification:** We demonstrate that for any two half-spaces in $\mathbb{Z}$, the compiler identifies unsatisfiable systems and collapses the resulting binary into a `ret void` or no-op sequence.
- **Toolchain Transparency:** To ensure reproducibility, we generate automated links to the *Compiler Explorer* (Godbolt). This allows reviewers to interactively verify the structural pruning across diverse toolchain versions, providing an empirical witness to the library's "Axiomatic Systems Programming" paradigm.

By restricting the initial implementation to exact number systems, we provide an airtight proof of concept for structuralist optimization, establishing the foundation for handling more complex transcendental continuums in future work.

# Chapter 15

# Discussion

**Symmetry with Standard Ranges:** While the use of `operator»` for Kleisli composition may appear novel, it serves as a direct conceptual mirror to the C++20 Ranges `operator|`. Where Ranges utilize the pipe to model the directional flow of data through non-owning views, `dedekind` utilizes the Highway to model the flow of monadic and comonadic context through morphisms. This "syntactic scent" aligns with the modern C++ idiom of declarative pipelines, transforming the "callback hell" of nested monadic binds into a linear, scannable vector of computation. By adopting a precedence and associativity model that echoes the Standard Library's approach to data streams, we ensure that categorical errors become visually discordant, significantly reducing the cognitive load for developers accustomed to the Ranges API.

**The Decidability Gap and Axiomatic Trust:** A common critique of embedding formal methods in C++ is the lack of native support for dependent types, which prevents the compiler from verifying the internal logic of a morphism. We address this via *Instrumented Model Validation*. While C++ Concepts verify the *Structural Identity* (the existence and signature of $\phi, \eta, \mu$), our CI/CD pipeline acts as an independent auditor that provides a *Runtime Witness* for the algebraic laws. By combining compile-time constraint satisfaction with property-based execution, we ensure that the "Axiomatic Atlas" remains a faithful representation of mathematical truth. Furthermore, we posit that the successful structural pruning performed by the LLVM backend serves as an empirical verification: if the categorical laws were not preserved, the compiler would be unable to collapse contradictory symbolic predicates into zero-overhead machine code.

**The Pragmatic Displacement of Symbolic Environments:** While "pure" functional languages offer higher-order proof-search, the practical baseline for symbolic computing remains high-level interpreted environments like Python. In these systems, mathematical invariants are treated as opaque runtime implementation details, leading to a profound "semantic gap" between intent and execution. `dedekind` addresses this not by seeking the "perfect" decidability of a theorem prover, but by providing a "sufficiently verified" structuralist alternative. By hoisting categorical requirements into C++23 Concepts, we provide a level of formal rigor that is absent in interpreted backends, while achieving the mechanical sympathy required to outperform them by orders of magnitude. We prioritize the "scent" of the implementation over the "perfection" of the proof, ensuring that the library remains a viable tool for high-performance Axiomatic Systems Programming rather than a purely theoretical exercise.

However, reifying a formal ontology in a statically-typed **systems** language presents significant challenges compared to "pure" functional languages like Haskell [49] or Agda [50], which have native support for higher-kinded types, dependent types and first-class functions. Challenges encountered include:

- **Template Template Parameters**: C++ is notoriously bad at true Category Theory because it lacks higher-kinded Types (HKTs). As a result `Functor<F>` or `Monad<F>` are defined in terms of `template template`, which have significant edge cases (e.g., they don't play well with aliases or multiple template arguments).

- **The Lambda Hurdle (Opaqueness)**: Unlike the first-class, inspectable functions in the ML family, C++ lambdas are unique, anonymous types. They do not inherently expose their domain or codomain to the type system. A morphism's domain and codomain must be inferred at the call site or explicitly provided by the programmer. Although if used as *Non-Type Template Parameters* (NTTP), they can be "inspected" them via `concept` checks or by wrapping them in a type that carries its own signature metadata.

- **Decidability vs. Instantiation**: While dependent-type systems use proof-search, C++ relies on template instantiation. This introduces the risk of "Late Detection," where a mathematical violation is only discovered when a set is materialized, rather than when it is defined. This can be mitigated by forcing evaluations into `consteval` and `constexpr` contexts.

- **The Performance-Rigidity Tradeoff**: High-level abstractions in C++ often incur a "template tax" in compile times. We utilize C++23 **Module Partitions** to create a directed acyclic graph of truth, caching structural invariants to ensure that formal rigor does not collapse the build pipeline.

- **The Specialization Constraint**: Unlike the unified pattern matching of functional languages, C++ relies on Partial Template Specialization to resolve structural properties. This creates a "Rigidity Challenge": the compiler cannot readily deduce that a property of $A$ also applies to $A \cup B$ without explicit boilerplate. We address this by using `concept ... requires` as logical compile-time predicate, thereby simulating the polymorphic reach of a true dependent-type system.

- **The Precedence-Associativity Constraint**: A significant friction point in embedding categorical notation within the `C++` grammar is the language's fixed set of operator candidates and their immutable precedence rules. While `operator»=` is the "nominal" choice for Monadic Bind ($\gg=$), the ISO standard defines all compound assignment operators with right-to-left associativity. Consequently, a standard monadic chain such as `m »= f »= g` is parsed as `m »= (f »= g)`, which violates the directional vector of the Kleisli triple.

  Rather than yielding to this syntactic prejudice, `dedekind` introduces the **Highway Notation** (`operator»`). This is not a mere workaround for the "Bind" associativity; it is a deliberate architectural choice to prioritize the left-to-right "push" of the structural pipeline. By utilizing the left-associativity of the shift operator, the Highway reifies the categorical flow as a first-class visual entity. This ensures that the code's "scent" remains isomorphic to the mathematical intent, even when the language's formal grammar resists the composition.

# Chapter 16

# Conclusion

This paper has demonstrated the feasibility of embedding formal mathematical structuralism within the `C++23` type system through the `dedekind` library. By transitioning from the traditional object-oriented paradigm of nominal inheritance toward a model of *structural transparency*, we have illustrated how the modern compiler can be utilized as a deterministic constraint satisfaction engine. This approach allows for the high-level representation of complex mathematical species—such as intensional set-builder notation and the Dedekind-complete continuum—without the runtime performance penalties historically associated with abstract symbolic modeling.

Our investigation suggests that the `Clang/LLVM` toolchain, while not a dedicated theorem prover, is capable of implementing a "rethought" set theory where categorical axioms provide the semantic context for aggressive machine-level optimization. By hoisting mereological relations into the type signature, we enable the backend to perform *structural pruning*: collapsing logical contradictions and optimizing intersections into zero-instruction sequences that maintain the efficiency of hand-tuned assembly.

The stratification of these formalisms into verified module partitions provides a scalable pathway for managing the "template tax," ensuring that the rigors of formal logic do not destabilize the build pipeline. By embedding the construction of the Continuum directly into the translation unit, we have moved toward a nascent paradigm of **Axiomatic Systems Programming**. In this model, the build process acts as a physical guardrail for mathematical truth, ensuring that high-level algebraic invariants and low-level hardware sympathy are no longer dualities, but a unified, verified reality.

Future work will extend this mereological framework to the upcoming linear algebra capabilities of the `C++` standard library [51], expressly facilitating exact computations across semimodules and *reduced* (booleans, `char`, `unsigned int`) or *extended* (complex, dual, quaternions) number systems. By anchoring these structures in the `dedekind` atlas, we aim to provide a unified algebraic substrate that transcends the binary-limitations of traditional hardware types, enabling a truly symbolic systems programming environment.

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